Journal Name

Crossmark

RECEIVED
dd Month yyyy

REVISED
dd Month yyyy

PAPER

Information Budgets of Trajectory Health Indicators for Remaining Useful Life Prediction

Anonymous authors

Keywords: information budget, health indicator, remaining useful life, information theory, monotonicity, information bottleneck, first hitting time, prognostics

Abstract

Health indicators (HIs) for remaining useful life (RUL) prediction are usually constructed by optimising geometric criteria, such as monotonicity, trendability and prognosability, or by training an encoder. Neither approach states how much information about the RUL an HI can retain. We model HI construction as a Markov chain from latent degradation to sensors to HI, and bound the mean squared error of any RUL predictor below by the entropy power of the RUL given the HI. The bound defines an information efficiency, converts the error of any predictor into a certified lower bound on the retained information, and accounts for the age, which every predictor knows. We show that a pointwise squeeze of the HI loses state information while monotonicity saturates; that, for Wiener degradation with an unobserved inspection state, the error splits into an aleatoric floor and an epistemic term that grows with the information lost; that a static scalar linear HI inflates the bound by a factor determined by the angle between its weights and the optimal weights, which rotate from velocity to level as the remaining life shrinks, so that no static scalar linear HI is optimal over the whole life; and that a regime-blind read-out has an error floor. Simulations on exactly solvable models agree with these results. We further show that an integrating encoder accumulates error that an observation correction removes. Experiments on batteries, bearings and turbofan engines support these consequences but show no learned encoder to be superior to simple indicators when all are used with the same age-fused predictor. On turbofan engines and lithium-ion cells with 77 to 260 units, the age-conditional certificate is informative and passes a negative control, and the monotonicity of a squeezed indicator saturates at the predicted value while its certified information vanishes. The loss predicted for a static scalar linear indicator is small for engines and for one cell collection but reaches 9 to 18 percent of the error for the other.

1 Introduction

Prognostics and health management (PHM) predicts the remaining useful life (RUL) of components such as batteries, bearings and aero-engines, and underpins condition-based maintenance [1, 2, 3, 4]. Methods are conventionally divided into physics-based, data-driven and hybrid [3, 5], and recent work is dominated by deep networks that map sensor histories to the RUL end to end [6, 7]. An established alternative compresses the monitoring data into a scalar *health indicator* (HI) and extrapolates it to a failure threshold D_{fail}. An HI can be inspected by an engineer, the threshold gives the end of life (EOL) a physical meaning, and, when the HI is modelled as a stochastic process, the RUL is a first hitting time (FHT) with a known distribution [8, 9, 1]. Recent studies in this journal follow the same line [10, 11, 12], within the wider scope of reliability science [13, 14].

HIs are usually selected by *geometric* criteria. Coble and Hines [15] formalised monotonicity (balance of positive and negative increments), prognosability (spread of end-of-life values relative to the range travelled) and trendability (agreement of the functional form across units) and optimised them with a genetic algorithm. Variants are common: trendability is often implemented as the correlation with time [16, 17], and a criterion based on ranking mutual information has been proposed for building a monotonic HI [18]. Coble and Hines themselves note that the no-self-healing assumption behind monotonicity fails for batteries [15]. HIs may also be learned by recurrent networks [19], autoencoders and their regularised variants [20, 21], or regressions on engineered features [22]. In both cases an HI is judged by its shape or by the accuracy of the

predictor built on it [23]. Neither judgement indicates how much information about the RUL the HI *could* have preserved, and neither separates the quality of the HI from that of the predictor.

Probabilistic RUL prediction from an HI is mostly built on Wiener degradation, whose first passage time is inverse Gaussian [24, 25, 26], with measurement error [8, 27], random unit-specific drift [28, 29], nonlinear drift [30] or threshold regression [9], applied to batteries [27], bearings [31] and highly reliable products [32] (reviews: [33, 1]). Two features are important for what follows: the drift varies between units, and the degradation level at inspection is inferred from the sensors rather than observed. The RUL uncertainty then has an aleatoric part (future randomness) and an epistemic part (incomplete knowledge of state and parameters) [34, 35, 36, 37]; deep ensembles are a common way to represent the latter [38].

Information theory [39, 40] enters learning mainly through the information bottleneck [41], which has variational forms [42, 43], links to invariance [44], nonlinear extensions [45] and critics [46]. Estimating entropy and mutual information from samples is difficult: low-dimensional estimators exist [47, 48], but their bias and variance are large in general [49], and a distribution-free, high-confidence lower bound on mutual information cannot exceed order $\ln N$ for N samples [50]. Limits on estimator accuracy are usually stated through Cramér–Rao bounds, in their Bayesian form due to van Trees [51, 52, 53]. The entropy-power bound used in this paper is the information-theoretic counterpart of these bounds and holds for arbitrary densities [40] (Remark 1).

The literature thus offers geometric and accuracy-based ways of judging an HI, a mature probabilistic theory of the RUL *given* an HI, and general information-theoretic tools. In the sources we searched (bibliographic databases and the references above; this is not an exhaustive survey) we did not find a theory that connects them, that is, a bound on the RUL error that depends on the HI only through the information it retains, a statement of what the geometric criteria do and do not measure in these terms, and an account of what a *scalar, static* indicator loses when the degradation velocity varies in time or the sensors respond to the operating regime. A companion study [54] analysed what a static, per-snapshot projection of a multichannel signal retains about the degradation state. Here we analyse the HI *trajectory*, with its level, velocity, between-unit variability and operating regime. Two further studies by the same authors address how many run-to-failure trajectories [55] and which charging-phase features [56] are needed; the present paper addresses how much *information* an HI can at best retain. Figure 1 shows the setting and where each result applies.

The contributions of the paper are:

1. an **information budget**: an entropy-power bound on the RUL error in terms of the mutual information $I(T_t; H_{1:t})$ between the RUL and the HI, an efficiency η_{info} that scores an HI before a predictor is trained, a certificate that turns any held-out error into a lower bound on retained information, age-aware versions of both, and a state–kinematic decomposition (Section 2);

2. **why geometric criteria do not measure information**: a pointwise squeeze loses state information while monotonicity saturates at a value set by the template (Section 3);

3. a **physics-constrained budget** for Wiener degradation with an unobserved inspection state: an HI-independent aleatoric floor plus an epistemic term amplified by information loss, and an exact non-Gaussianity gap (Section 4);

4. an **angular loss law**: a static scalar linear HI inflates the bound by $1 + \rho \sin^2 \psi$, where ψ is the angle between its weights and the optimal weights and ρ a signal-to-noise ratio, because the optimal weights rotate from velocity to level; no static scalar linear HI is therefore optimal over the whole life, and a regime-blind read-out has an error floor (Section 5);

5. **consequences for estimators**: an exact path-space bottleneck objective, and a proof that an integrating encoder accumulates error that an observation correction removes (Section 6), examined on batteries, bearings and turbofan engines with a conservative protocol (Section 7), and the predictions of the theory are tested on turbofan engines (100 to 260 units per subset) and lithium-ion cells (77 and 169 units) (Section 8).

The ingredients are classical: the data processing inequality, the chain rules, the maximum-entropy property of the Gaussian and the entropy-power bound [40]; the inverse-Gaussian law [25]; Gaussian conditioning [57]; the Kalman filter [58, 59]; and the variational bottleneck bound [42]. What is new is their use as a budget for the HI-construction step, and the specific results listed

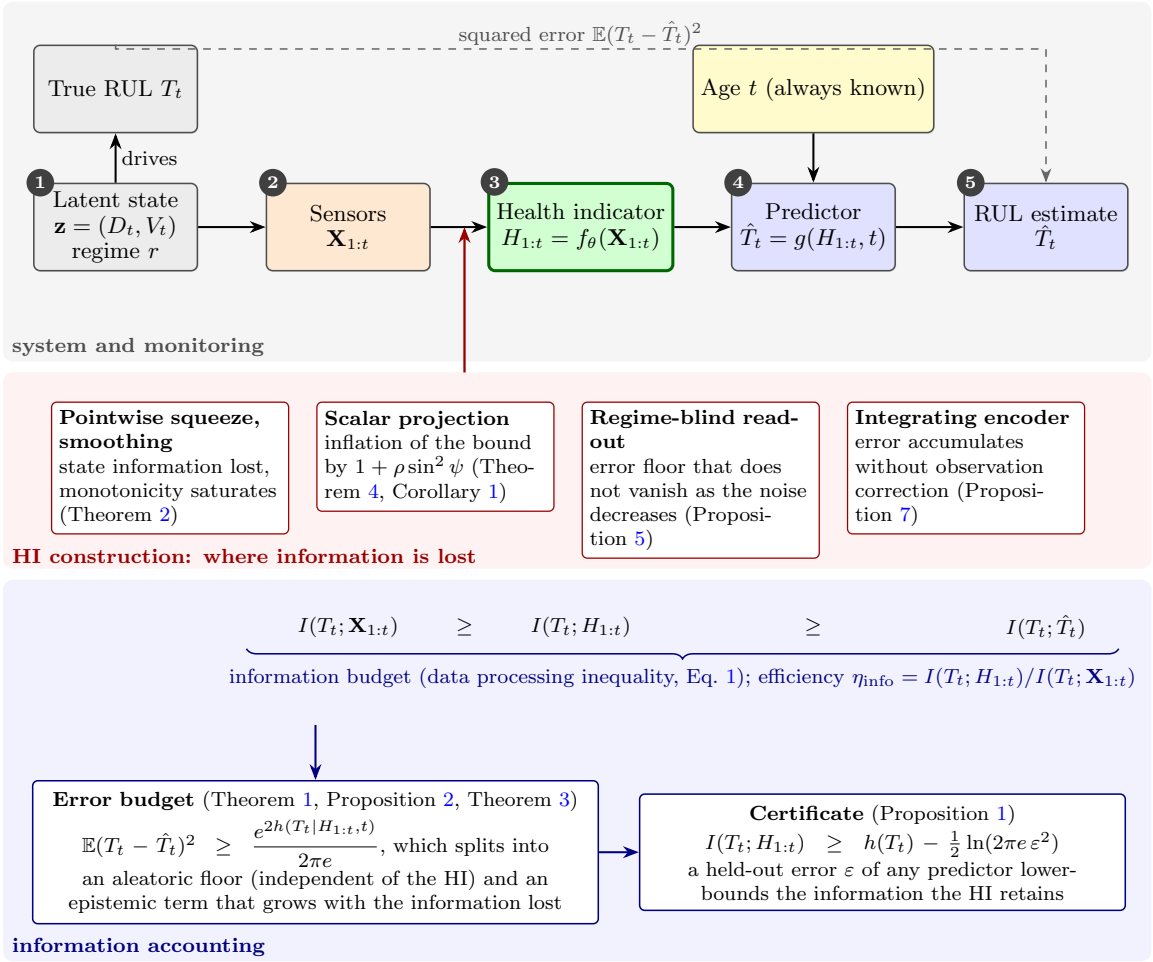


Figure 1. Framework of the paper. The HI is constructed from the sensors, which are driven by the same latent state as the true RUL, and the predictor uses the HI and the age. The data processing inequality gives an information budget; the red boxes list the operations of HI construction whose information loss is quantified.

above. Each proof states which classical facts it uses and where they can be found, and Sections 3 to 5 each end with a numerical check on an exactly solvable model.

2 Information budget of a health indicator

2.1 Setting

Let $T_t > 0$ be the true RUL at inspection time (age) t, $\mathbf{X}_{1:t} \in \mathbb{R}^{d\times t}$ the accumulated history of d sensors, and $H_{1:t} = f_\theta(\mathbf{X}_{1:t})$ the HI trajectory produced by a map f_θ with parameters θ. Throughout, $h(\cdot)$ is differential entropy, $I(\cdot\,;\cdot)$ is mutual information (both in nats), $\mathbb{E}$ and Var are expectation and variance, and $\mathcal{N}(\mu, \sigma^2)$ is the normal distribution. Any RUL estimator $\hat{T}_t = g(H_{1:t})$ obeys the Markov chain $T_t \to \mathbf{X}_{1:t} \to H_{1:t} \to \hat{T}_t$, because the latent degradation state drives both T_t and $\mathbf{X}_{1:t}$. The data processing inequality (DPI) [40, Sect. 2.8] gives

$$I(T_t; \mathbf{X}_{1:t}) \geq I(T_t; H_{1:t}) \geq I(T_t; \hat{T}_t), \tag{1}$$

with equality in the first inequality if and only if $H_{1:t}$ is a sufficient statistic of $\mathbf{X}_{1:t}$ for T_t [40, Sect. 2.9].

Theorem 1 (Entropy-power bound on RUL error)**.** *Assume that $h(T_t)$ and $I(T_t; \mathbf{X}_{1:t})$ are finite. For every estimator $\hat{T}_t = g(H_{1:t})$ with finite second moment, the mean squared error (MSE) satisfies*

$$\mathbb{E}\big[(T_t - \hat{T}_t)^2\big] \;\geq\; \mathcal{B}_{\text{info}}(H_{1:t}) := \frac{1}{2\pi e}\exp\big(2h(T_t \mid H_{1:t})\big) = \frac{1}{2\pi e}\exp\big(2h(T_t) - 2I(T_t; H_{1:t})\big). \tag{2}$$

Moreover $\mathcal{B}_{\text{info}}(H_{1:t}) \geq \mathcal{B}_{\min} := \frac{1}{2\pi e} e^{2h(T_t|\mathbf{X}_{1:t})}$ and

$$\mathcal{B}_{\text{info}}(H_{1:t})/\mathcal{B}_{\min} = \exp\big(2\,(1 - \eta_{\text{info}}(H_{1:t}))\, I(T_t; \mathbf{X}_{1:t})\big), \qquad \eta_{\text{info}}(H_{1:t}) := \frac{I(T_t; H_{1:t})}{I(T_t; \mathbf{X}_{1:t})} \in [0, 1]. \tag{3}$$

Proof. Let $E = T_t - g(H_{1:t})$. Differential entropy is translation invariant [40, Sect. 8.6], so $h(E \mid H_{1:t}) = h(T_t \mid H_{1:t})$ ($g(H_{1:t})$ is constant given $H_{1:t}$). Conditioning does not increase entropy [40, Sect. 2.6], so $h(E \mid H_{1:t}) \le h(E)$, and the Gaussian maximises entropy at fixed second moment [40, Sect. 8.6], so $h(E) \le \frac{1}{2}\ln(2\pi e\,\mathbb{E}[E^2])$. Combining the three relations gives $\mathbb{E}[E^2] \ge \frac{1}{2\pi e} e^{2h(T_t|H_{1:t})}$. The identity $h(T_t \mid H_{1:t}) = h(T_t) - I(T_t; H_{1:t})$ (definition of mutual information) gives the second form. By (1), $h(T_t \mid H_{1:t}) \ge h(T_t \mid \mathbf{X}_{1:t})$, and the ratio follows from $h(T_t \mid H_{1:t}) - h(T_t \mid \mathbf{X}_{1:t}) = I(T_t; \mathbf{X}_{1:t}) - I(T_t; H_{1:t})$. □

Proposition 1 (Certified information)**.** *Under the assumptions of Theorem 1, any estimator $\hat{T}_t = g(H_{1:t})$ with mean squared error ε^2 certifies*

$$I(T_t; H_{1:t}) \;\ge\; h(T_t) - \tfrac{1}{2}\ln\bigl(2\pi e\,\varepsilon^2\bigr). \tag{4}$$

Proof. Rearrange $\varepsilon^2 \ge \frac{1}{2\pi e}\exp(2h(T_t) - 2I(T_t; H_{1:t}))$ from Theorem 1. □

Proposition 2 (Age-aware bound)**.** *The age t is always available to a predictor. For every estimator $\hat{T}_t = g(H_{1:t}, t)$, $\mathbb{E}[(T_t - \hat{T}_t)^2] \ge \frac{1}{2\pi e}\exp\bigl(2h(T_t \mid H_{1:t}, t)\bigr)$, and the information that the HI adds to the age is*

$$I(T_t; H_{1:t}, t) = I(T_t; t) + I(T_t; H_{1:t} \mid t), \qquad I(T_t; H_{1:t} \mid t) \ge \mathbb{E}_t\, h(T_t \mid t) - \tfrac{1}{2}\ln\bigl(2\pi e\,\varepsilon^2\bigr), \tag{5}$$

where ε^2 is the MSE of any estimator that uses $H_{1:t}$ and t.

Proof. Apply Theorem 1 conditionally on t and average. The chain rule of mutual information [40, Sect. 2.5] gives the decomposition. For the inequality, Theorem 1 conditionally on t yields $I(T_t; H_{1:t} \mid t = \tau) \ge h(T_t \mid t = \tau) - \frac{1}{2}\ln(2\pi e\,\varepsilon_\tau^2)$ with ε_τ^2 the conditional MSE; averaging over τ and using Jensen's inequality for the concave logarithm [40, Sect. 2.6], $\mathbb{E}_\tau \ln \varepsilon_\tau^2 \le \ln \mathbb{E}_\tau \varepsilon_\tau^2 = \ln \varepsilon^2$, gives the stated bound. □

Remark 1 (Relation to the Bayesian Cramér–Rao bound)**.** *Let $J(h)$ be the Fisher information, with respect to location, of the posterior density $p(T_t \mid H_{1:t} = h)$. Under the regularity conditions of the van Trees inequality [51, 52] every estimator satisfies $\mathbb{E}[(T_t - \hat{T}_t)^2] \ge 1/\mathbb{E}_H[J(H)]$. Stam's inequality [60] states that $e^{2h(T_t|H=h)}/(2\pi e) \ge 1/J(h)$ for every h. Taking the geometric mean over h and using that the geometric mean of $1/J$ is at least the reciprocal of the arithmetic mean of J,*

$$\mathcal{B}_{\text{info}}(H_{1:t}) = \exp\Bigl(\mathbb{E}_H \ln \frac{e^{2h(T_t|H=h)}}{2\pi e}\Bigr) \;\ge\; \exp\Bigl(\mathbb{E}_H \ln \frac{1}{J(H)}\Bigr) \;\ge\; \frac{1}{\mathbb{E}_H[J(H)]}, \tag{6}$$

so, wherever the Bayesian Cramér–Rao bound applies, the entropy-power bound is not weaker, and it requires no differentiability of the densities. For posterior filtering the Fisher-information route is developed in [53].

Theorem 1 is the classical entropy-power bound on the error of an estimator [40, Sect. 8.6], used here to turn HI construction into an information budget. It is tight only if the posterior of T_t is Gaussian and is otherwise conservative (Section 4.1); $\mathcal{B}_{\text{info}}$ equals $\mathcal{B}_{\min}$ exactly when $\eta_{\text{info}} = 1$.

Proposition 2 shows that the relevant baseline for an HI is the RUL given the age, not the marginal distribution of the RUL. A sensor-free predictor that uses only the age is therefore a necessary reference (Appendix A). Proposition 1 gives a quantity that can be computed from real data, since it needs only the one-dimensional entropy $h(T_t)$ and the held-out MSE of a predictor. It is a monotone transform of the MSE and therefore ranks HIs in the same order as the MSE; it is the information-theoretic reading of the error, not independent evidence, and it is vacuous (negative) when the predictor is poor. If the predictor uses the age, the certificate applies to the HI trajectory together with its length t, that is, it certifies $I(T_t; H_{1:t})$, which already contains the age. The information that the HI adds to the age, $I(T_t; H_{1:t} \mid t)$, is certified only by Proposition 2, which requires the entropy $h(T_t \mid t)$ of the RUL given the age. The difference between the certificate of an HI-and-age predictor and that of an age-only predictor is a difference of two lower bounds and is not a lower bound on the information that the HI adds. The efficiency η_{info} could in principle be estimated before any predictor is trained, but mutual-information estimates are unreliable on small data sets [49, 50] (Appendix C).

2.2 Trajectory decomposition

Proposition 3 (State–kinematic decomposition)**.** *Let $\Delta H_k = H_k - H_{k-1}$ denote the increments of the HI. The map $H_{1:t} \mapsto (H_t, \Delta H_{2:t})$ is a linear bijection, hence*

$$I(T_t; H_{1:t}) = \underbrace{I(T_t; H_t)}_{\mathcal{I}_{\text{state}}(t)} + \underbrace{I(T_t; \Delta H_{2:t} \mid H_t)}_{\mathcal{I}_{\text{kin}}(t)}. \tag{7}$$

Proof. The map has a unit-determinant matrix, so $h(\mathbf{AY}) = h(\mathbf{Y}) + \ln|\det \mathbf{A}|$ [40, Sect. 8.6] and mutual information is unchanged ($H_k = H_t - \sum_{j=k+1}^{t} \Delta H_j$ gives the inverse). The identity is then the chain rule of mutual information [40, Sect. 2.5]. □

Remark 2. *The decomposition is an identity; its content lies in which operations remove which part. If the drift is constant, H_t/t is a sufficient statistic for the drift, and $\mathcal{I}_{\text{kin}}$ about the drift is zero. Kinematic information is non-trivial when the velocity varies in time, for example under a state-dependent drift or a random-walk velocity, as in late-life acceleration.*

3 Geometric criteria do not measure information

The geometric criteria of Coble and Hines [15] are computed from the shape of the HI. Monotonicity is $\text{Mon}(H_{1:t}) = |\sum_{k=1}^{t-1} \text{sgn}(H_{k+1} - H_k)|/(t-1)$, the balance of positive and negative increments. Prognosability is $\text{Pro} = \exp(-\text{std}(H_{\text{end}})/\text{mean}|H_{\text{end}} - H_{\text{start}}|)$, where the standard deviation and the mean are taken over units, H_{start} and H_{end} being the first and last values of each unit. In the benchmarks, Tre is the absolute correlation of the HI with time, as in [16] (and, under the name correlation, in [17]); it differs from the trendability of Coble and Hines, which measures the agreement of the functional form across units [15].

A strictly monotone bijection ϱ satisfies $I(T_t; \varrho(H_{1:t})) = I(T_t; H_{1:t})$, so information can be lost only through non-injective operations or noise. We therefore consider $\tilde{H}_{1:t} = \mathcal{S}(H_{1:t}, \xi)$, where ξ is independent of everything else. We call $\mathcal{S}$ a *squeeze* when it contracts the HI towards a template trajectory.

Theorem 2 (Information inflation under squeezing)**.** *(a) For any such $\mathcal{S}$, $T_t \to H_{1:t} \to \tilde{H}_{1:t}$ is Markov, so $I(T_t; \tilde{H}_{1:t}) \le I(T_t; H_{1:t})$ and*

$$\mathcal{B}_{\text{info}}(\tilde{H}_{1:t})/\mathcal{B}_{\text{info}}(H_{1:t}) = \exp\big(2[I(T_t; H_{1:t}) - I(T_t; \tilde{H}_{1:t})]\big) \ge 1. \tag{8}$$

(b) If $\mathcal{S}$ acts pointwise in time, $\tilde{H}_k = \phi_k(H_k, \xi_k)$, then

$$\begin{gathered} \ln \frac{\mathcal{B}_{\text{info}}(\tilde{H}_{1:t})}{\mathcal{B}_{\text{info}}(H_{1:t})} = 2\big(\Delta\mathcal{I}_{\text{state}} + \Delta\mathcal{I}_{\text{kin}}\big), \\ \Delta\mathcal{I}_{\text{state}} := \mathcal{I}_{\text{state}}(H) - \mathcal{I}_{\text{state}}(\tilde{H}) \ge 0, \quad \Delta\mathcal{I}_{\text{kin}} := \mathcal{I}_{\text{kin}}(H) - \mathcal{I}_{\text{kin}}(\tilde{H}), \end{gathered} \tag{9}$$

and $\Delta\mathcal{I}_{\text{kin}}$ has no definite sign in general: kinematic information alone does not determine the inflation.

(c) Let the HI be the latent degradation level D_k observed with Gaussian noise, $H_k = D_k + \xi_k^{(1)}$, where D_k is linear-Gaussian, and let the squeeze with strength $\gamma \in [0, 1)$ and template (mean trajectory) $m_k = \mathbb{E}[D_k]$ be

$$\tilde{H}_k^{\gamma} = m_k + (1-\gamma)(H_k - m_k) + \xi_k^{(2)}, \qquad \xi_k^{(2)} \sim \mathcal{N}(0, \sigma_2^2) \text{ independent and identically distributed (i.i.d.)} \tag{10}$$

For every linear functional ϑ of the latent state (for example the current velocity), write $\mathcal{I}^{\vartheta}_{\text{state}}(\gamma) = I(\vartheta; \tilde{H}_t^{\gamma})$, $\mathcal{I}^{\vartheta}_{\text{kin}}(\gamma) = I(\vartheta; \Delta\tilde{H}^{\gamma}_{2:t} \mid \tilde{H}_t^{\gamma})$ and $\mathcal{I}^{\vartheta}_{\text{tot}}(\gamma) = I(\vartheta; \tilde{H}^{\gamma}_{1:t})$. Then $\mathcal{I}^{\vartheta}_{\text{tot}}$ and $\mathcal{I}^{\vartheta}_{\text{state}}$ are non-increasing in γ, and

$$0 \le \mathcal{I}^{\vartheta}_{\text{state}}(\gamma),\ \mathcal{I}^{\vartheta}_{\text{kin}}(\gamma) \le \mathcal{I}^{\vartheta}_{\text{tot}}(\gamma) \xrightarrow[\gamma \to 1]{} 0, \tag{11}$$

so both components vanish, although $\mathcal{I}^{\vartheta}_{\text{kin}}$ need not be monotone. If the template has constant increment Δm, then as $t \to \infty$ and $\gamma \to 1$,

$$\text{Mon}(\tilde{H}^{\gamma}_{1:t}) \to \text{Mon}_{\infty} = 2\Phi_{\mathcal{N}}\big(\Delta m/(\sqrt{2}\,\sigma_2)\big) - 1, \tag{12}$$

where $\Phi_{\mathcal{N}}$ is the standard normal distribution function; the limit does not depend on the unit's own degradation.

Table 1. Squeezing experiment (mean over three seeds; ±: standard deviation). Information about V_t in nats, errors in cycles.

γ	Mon	$\mathcal{I}_{\text{state}}$ [nats]	$\mathcal{I}_{\text{kin}}$ [nats]	η_{info}	$\sqrt{\mathcal{B}_{\text{info}}}$ [cycles]	Bayes RMSE [cycles]	Empirical RMSE [cycles]
0.000	0.496	0.789	0.092	1.00	6.67	8.04	15.0±9.8
0.500	0.766	0.779	0.092	0.99	6.81	8.91	14.9±9.4
0.900	0.975	0.567	0.188	0.85	7.38	9.27	15.3±9.2
0.970	0.981	0.159	0.396	0.63	9.01	10.99	17.1±9.2
0.990	0.981	0.022	0.200	0.25	12.87	15.20	21.2±8.9
0.995	0.982	0.006	0.070	0.09	14.88	17.83	23.2±8.9

Proof. (a) $\tilde{H}$ is a randomised function of H only, so the Markov property holds and the DPI gives the first claim; the ratio follows from the definition of $\mathcal{B}_{\text{info}}$. (b) Apply Proposition 3 to H and to $\tilde{H}$ and take the difference in (a). Because $\tilde{H}_t = \phi_t(H_t, \xi_t)$ is a randomised function of H_t alone, the DPI gives $I(T_t; \tilde{H}_t) \le I(T_t; H_t)$, that is, $\Delta\mathcal{I}_{\text{state}} \ge 0$. For non-pointwise operators (for example smoothing over history) even this sign can fail. (c) With $\lambda = 1 - \gamma > 0$, $\tilde{H}^\gamma$ is an invertible affine image of $(H - m) + \xi^{(2)}/\lambda$, that is, of H observed through additive Gaussian noise of variance σ_2^2/λ^2. For $\gamma_2 > \gamma_1$ the noise of the second channel is the first plus independent Gaussian noise, so the DPI [40, Sect. 2.8] gives monotonicity of $\mathcal{I}^\vartheta_{\text{tot}}$ and, applying the same argument to the scalar observation at time t, of $\mathcal{I}^\vartheta_{\text{state}}$. Since $\mathcal{I}^\vartheta_{\text{state}} = I(\vartheta; \tilde{H}_t) \le I(\vartheta; \tilde{H}_{1:t})$ and $\mathcal{I}^\vartheta_{\text{kin}} = \mathcal{I}^\vartheta_{\text{tot}} - \mathcal{I}^\vartheta_{\text{state}} \le \mathcal{I}^\vartheta_{\text{tot}}$, both are dominated by $\mathcal{I}^\vartheta_{\text{tot}}$. As $\lambda \to 0$ the noise variance diverges, and the mutual information of a Gaussian channel, $\frac{1}{2}\ln(1 + \text{SNR})$ per scalar observation with signal-to-noise ratio SNR [40, Chap. 9], vanishes. For the limit of Mon, as $\lambda \to 0$ the increments converge to $\Delta m + \xi^{(2)}_{k+1} - \xi^{(2)}_k$, a stationary 1-dependent sequence. The indicators of a positive increment at odd k involve disjoint pairs of the $\xi^{(2)}$ and are therefore i.i.d., as are those at even k, so the strong law of large numbers [61, Chap. 2] applies to each subsequence and the fraction of positive increments tends to $p = \Phi_{\mathcal{N}}(\Delta m/(\sqrt{2}\sigma_2))$ and Mon $\to |2p - 1|$. □

Part (c) shows that the geometric score and the information content are decoupled. Once the squeeze is strong, Mon saturates at a value set by the template and the noise, while the information available for RUL prediction keeps falling and the bound $\mathcal{B}_{\text{info}}$ keeps rising. Part (b) separates two ways of losing information; in the numerical check below, the kinematic part even increases while the state part collapses.

3.1 Numerical check

The latent state is a local linear trend, $D_k = D_{k-1} + V_{k-1} + \sigma_B w_k$ and $V_k = V_{k-1} + \sigma_V \zeta_k$, with level D_k, velocity V_k, independent standard normal noises w_k, ζ_k and $V_0 \sim \mathcal{N}(\bar{v}, \sigma_{v0}^2)$. The raw HI is $H_k = D_k + \xi^{(1)}_k$ with measurement noise $\xi^{(1)}_k \sim \mathcal{N}(0, \sigma_1^2)$, and the squeeze is that of Theorem 2(c) with noise standard deviation σ_2. The parameters are $\bar{v} = 2$, $\sigma_{v0} = 0.25$, $\sigma_V = 0.03$, $\sigma_B = 0.5$, $\sigma_1 = 2$, $\sigma_2 = 0.6$, failure threshold $D_{\text{fail}} = 200$ and inspection at $t = 50$. The information about V_t is exact from the Kalman covariances [58, 59]. For the RUL, the Bayes-optimal estimator is the posterior mean from 300 exact posterior draws per unit, $h(T_t \mid \tilde{H})$ is estimated with the Vasicek estimator [47], and $\sqrt{\mathcal{B}_{\text{info}}}$ is the square root of the average of the pointwise entropy powers. Errors are reported as root mean squared error (RMSE); each setting uses 400 test units and three seeds.

Monotonicity saturates at the predicted value $\text{Mon}_\infty = 0.981$ for $\gamma \ge 0.9$, while the efficiency falls from 0.85 to 0.09 and the Bayes-optimal RMSE nearly doubles (9.3 to 17.8 cycles). The bound stays below the Bayes-optimal error in every run. The state information falls monotonically, whereas the kinematic information first rises and then falls, so the inflation at $\gamma = 0.9$ and 0.97 is driven by the loss of state information. The empirical RMSE varies strongly between seeds because of the heavy right tail of the RUL, and the posterior-based Bayes RMSE is the reliable summary.

4 A physics-constrained error budget

Assume that after inspection the latent degradation follows $dD_s = \mu\, ds + \sigma_B\, dW_s$, $\mu > 0$ unit-specific (the random-effect formulation of [28, 29]), with absorbing threshold D_{fail}. The prognostically relevant latent state is $\mathbf{z} = (D_t, \mu)$, where the inspection level D_t is *not* observed directly but only through $\mathbf{X}_{1:t}$. With $a = D_{\text{fail}} - D_t > 0$, the RUL is conditionally inverse-Gaussian [24, 26], $T_t \mid \mathbf{z} \sim \mathcal{IG}(m(\mathbf{z}), a^2/\sigma_B^2)$ with mean $m(\mathbf{z}) = a/\mu$ and variance $a\sigma_B^2/\mu^3$ [25], and T_t is independent of $\mathbf{X}_{1:t}$ (hence of $H_{1:t}$) given $\mathbf{z}$. Here $\mathcal{IG}(\text{mean}, \text{shape})$ denotes the inverse Gaussian distribution. Define the information efficiency about the mean RUL $m = m(\mathbf{z})$ as $\eta_m(H) := I(m; H_{1:t})/I(m; \mathbf{X}_{1:t}) \in [0, 1]$.

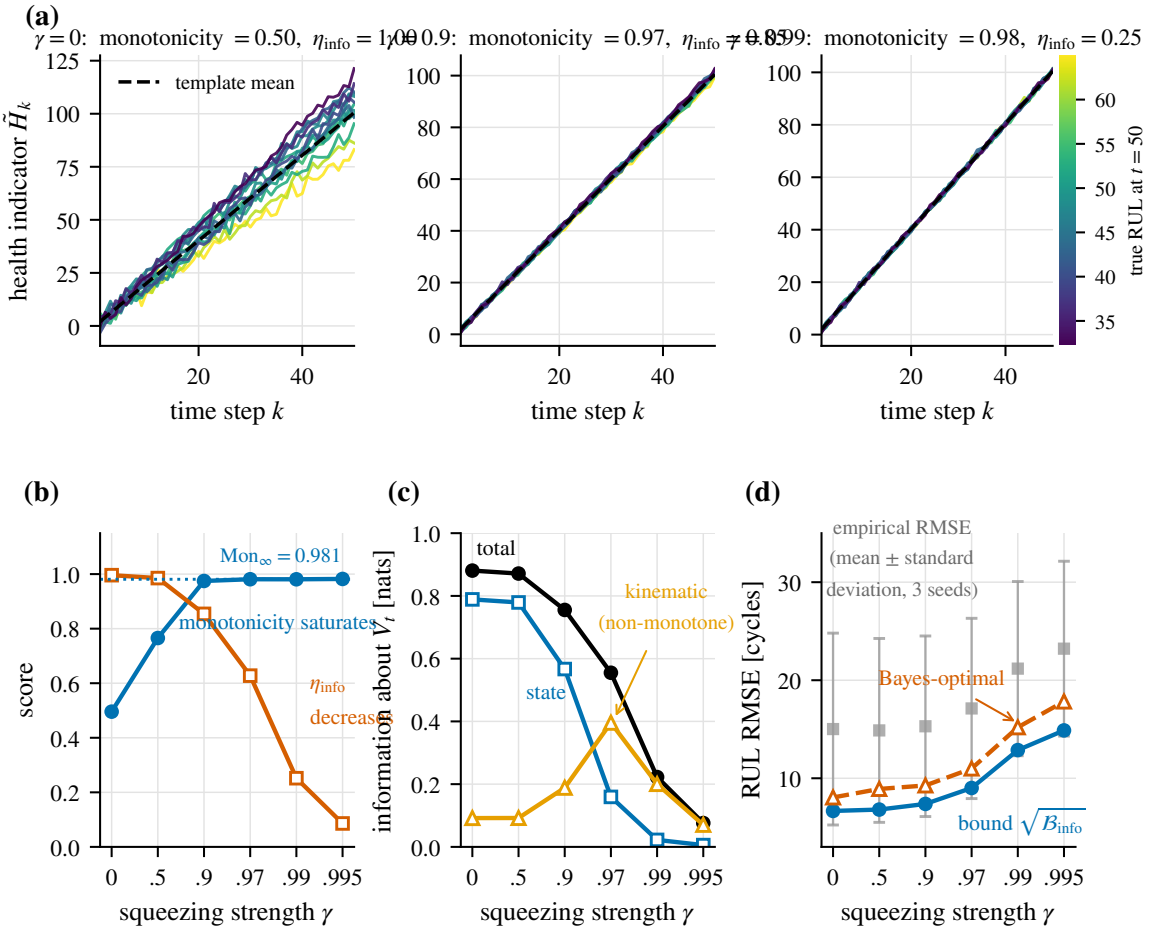


Figure 2. Squeezing experiment. (a) HI paths of 14 units, coloured by the true RUL. (b) Monotonicity and efficiency. (c) Information about V_t. (d) Bound and RMSE (error bars: standard deviation over seeds). The values of γ are equally spaced.

Theorem 3 (Physics-constrained bound)**.** *Assume $h(m \mid \mathbf{X}_{1:t})$ and $I(m; \mathbf{X}_{1:t})$ are finite. For any estimator $\hat{T}_t = g(H_{1:t})$ with finite second moment,*

$$\mathbb{E}[(T_t - \hat{T}_t)^2] \;\geq\; \underbrace{\mathbb{E}\left[\frac{a\,\sigma_B^2}{\mu^3}\right]}_{\mathcal{E}_{\text{aleatoric}}} + \underbrace{\mathbb{E}_H\left[\frac{e^{2h(m|H_{1:t}=h)}}{2\pi e}\right]}_{\mathcal{E}_{\text{epistemic}}}, \tag{13}$$

and the epistemic term obeys

$$\mathcal{E}_{\text{epistemic}} \;\geq\; \frac{1}{2\pi e}\exp\Big(2h(m \mid \mathbf{X}_{1:t}) + 2\big(1 - \eta_m(H)\big)\, I(m; \mathbf{X}_{1:t})\Big). \tag{14}$$

The aleatoric term does not depend on the HI.

Proof. Since $\mathbb{E}[T_t - m(\mathbf{z}) \mid \mathbf{z}, H_{1:t}] = 0$ and T_t is conditionally independent of $H_{1:t}$ given $\mathbf{z}$, the cross term vanishes and $\mathbb{E}[(T_t - \hat{T}_t)^2] = \mathbb{E}[\mathrm{Var}(T_t \mid \mathbf{z})] + \mathbb{E}[(m(\mathbf{z}) - \hat{T}_t)^2]$, the usual split of the squared error into an irreducible and an estimation part [57, Sect. 1.5.5]. The first term is $\mathbb{E}[a\sigma_B^2/\mu^3]$. For the second, apply Theorem 1 conditionally on $H_{1:t} = h$ to the target $m(\mathbf{z})$, which gives $\mathbb{E}[(m - \hat{T}_t)^2 \mid H = h] \geq e^{2h(m|H=h)}/(2\pi e)$, and average over h. For the second inequality, the exponential is convex, so (Jensen [40, Sect. 2.6]) $\mathbb{E}_H[e^{2h(m|H=h)}] \geq e^{2h(m|H)}$ with $h(m \mid H) = \mathbb{E}_H h(m \mid H = h)$; and $m \to \mathbf{X}_{1:t} \to H_{1:t}$ is Markov, so $h(m \mid H) = h(m) - I(m; H) = h(m \mid \mathbf{X}) + I(m; \mathbf{X}) - I(m; H) = h(m \mid \mathbf{X}) + (1 - \eta_m) I(m; \mathbf{X})$. □

Remark 3. *Conditioning on D_t and changing variables would give $h(m \mid H) = h(\mu \mid H) + \mathbb{E}\ln(a/\mu^2)$. In the constant-drift model, D_t/t is then a sufficient statistic for μ and the resulting bound no longer depends on the HI. The HI matters through the joint uncertainty about the level and the drift, which is why Theorem 3 is stated for the latent pair $\mathbf{z}$. The second inequality of the theorem (the Jensen form) is weaker than the first but depends on the HI only through η_m.*

Table 2. Physics-constrained bound. $\mathcal{E}_{\text{ale}}$, $\mathcal{E}_{\text{epi}}$: aleatoric and epistemic terms (cycles^2); I in nats; errors in cycles.

ς	η_m	$I(m;H)$ [nats]	$\mathcal{E}_{\text{ale}}$	$\mathcal{E}_{\text{epi}}$ (exact)	$\mathcal{E}_{\text{epi}}$ (pointwise)	$\mathcal{E}_{\text{epi}}$ (Jensen)	$\sqrt{\text{bound}}$ [cycles]	Bayes RMSE [cycles]
0	1.00	1.40	63.9	86.6	75.7	34.3	11.81	12.26
2	0.96	1.35	64.0	91.2	80.1	37.9	12.00	12.46
4	0.91	1.28	63.4	99.0	86.6	44.1	12.25	12.74
8	0.80	1.13	63.8	121.5	106.4	59.7	13.05	13.61
16	0.63	0.88	62.9	174.7	150.9	97.3	14.62	15.42
32	0.38	0.53	63.8	318.7	265.0	198.6	18.13	19.56

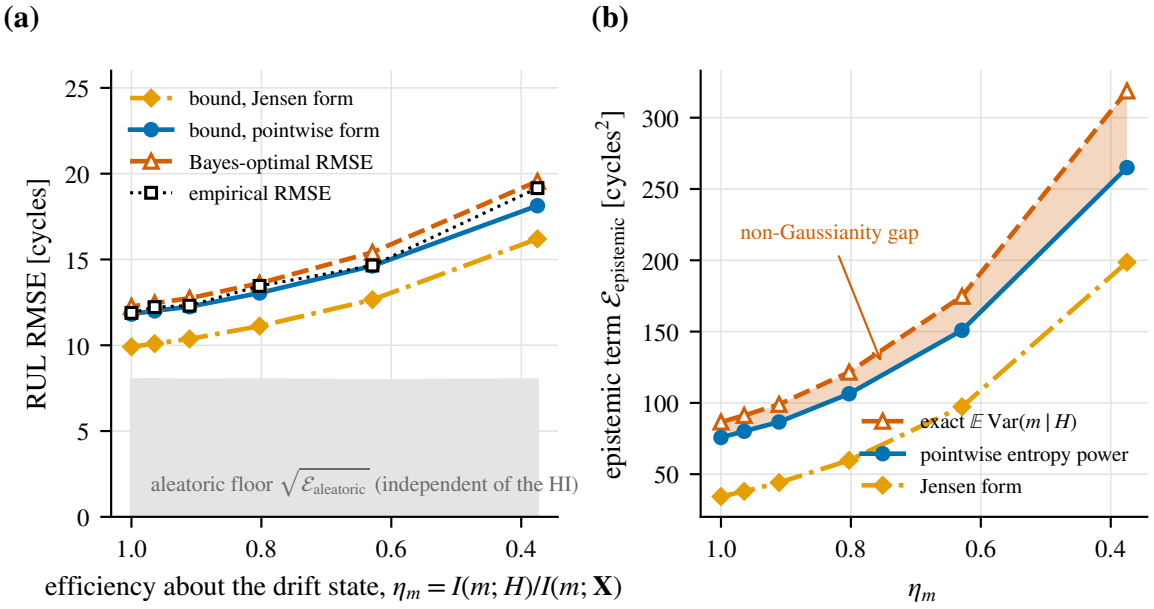


Figure 3. Physics-constrained bound as the HI loses information (decreasing η_m). (a) RMSE of the bounds and estimators. (b) Epistemic term; the shaded area is the non-Gaussianity gap.

Proposition 4 (Non-Gaussianity gap)**.** *Let $\hat{m}_H = \mathbb{E}[m \mid H]$ be the Bayes-optimal estimator of m and let $D_h := D_{\text{KL}}\big(p(m \mid H = h) \,\|\, \mathcal{N}(\mathbb{E}[m \mid h], \text{Var}(m \mid h))\big) \geq 0$ be the Kullback–Leibler (KL) divergence of the posterior from the Gaussian with the same mean and variance. Then*

$$\mathbb{E}\big[(m - \hat{m}_H)^2\big] - \mathbb{E}_H\left[\frac{e^{2h(m|H=h)}}{2\pi e}\right] = \mathbb{E}_H\Big[\,\text{Var}(m \mid H)\big(1 - e^{-2D_H}\big)\Big] \;\geq 0, \tag{15}$$

with equality if and only if $p(m \mid H)$ is Gaussian almost surely.

Proof. For a density p with mean μ_p and variance σ_p^2 and the Gaussian $q = \mathcal{N}(\mu_p, \sigma_p^2)$, $\int p \ln q = -\frac{1}{2}\ln(2\pi\sigma_p^2) - \frac{1}{2}$ because q depends on x only through the first two moments of p. Hence $D_{\text{KL}}(p\|q) = -h(p) + \frac{1}{2}\ln(2\pi e\sigma_p^2)$ (the identity behind the maximum-entropy property of the Gaussian [40, Sect. 8.6]), that is, $e^{2h(p)}/(2\pi e) = \sigma_p^2 e^{-2D_{\text{KL}}(p\|q)}$. Apply this to $p = p(m \mid H = h)$, use $\mathbb{E}[(m - \hat{m}_H)^2 \mid H = h] = \text{Var}(m \mid h)$, which holds because the posterior mean is the Bayes-optimal estimator under squared loss [57, Sect. 1.5.5], and average over h. The right-hand side vanishes if and only if $D_H = 0$ almost surely, that is, if and only if the posterior is Gaussian. □

4.1 Numerical check

The drift is $\mu \sim \mathcal{N}(1, 0.22^2)$, truncated to $\mu > 0.3$ and $a > 1$ (which keeps posterior sampling exact), with $\sigma_B = 0.85$, $D_{\text{fail}} = 100$ and $t = 50$. The raw sensor is $X_k = D_k + \xi_k$ with $\xi_k \sim \mathcal{N}(0, \sigma_\xi^2)$ and $\sigma_\xi = 1.5$. HIs with decreasing information are $H_k^\varsigma = X_k + \varsigma\zeta_k$ with independent standard normal ζ_k and $\varsigma \in \{0, 2, 4, 8, 16, 32\}$; each is a degraded version of X. The posterior of (D_t, μ) given H^ς is Gaussian (Kalman filter on (D, μ)), and $m = a/\mu$ is evaluated on 1500 posterior draws for each of 1500 test units; $h(m \mid H)$ and $h(m)$ are estimated with the Vasicek estimator. The Bayes-optimal error is $\mathbb{E}[\text{Var}(T_t \mid \mathbf{z})] + \mathbb{E}[\text{Var}(m \mid H)]$, and the empirical error against simulated first-passage times is also reported.

The aleatoric term is constant (≈ 63.8 cycles^2), whereas the epistemic term grows as η_m falls from 1 to 0.38. The pointwise bound stays below the Bayes-optimal error at every noise level

(RMSE 11.8 against 12.3 cycles for $\varsigma = 0$, and 18.1 against 19.6 for $\varsigma = 32$). The exact epistemic variance exceeds the pointwise value because the posterior of m is not Gaussian; the mean KL divergence in Proposition 4 is about 0.07 nats at $\varsigma = 0$. The marginal density (21) of Section 6, which mixes the inverse-Gaussian law over the Gaussian drift, agrees with numerical integration of the mixture to within 0.31%. The entropy estimates carry Monte Carlo error that we did not quantify, so gaps of a few per cent should not be over-interpreted.

5 Scalar indicators at the trajectory level

Theorem 3 concerns the information budget of an arbitrary HI. We now ask what a *scalar*, *static* HI loses when the degradation velocity varies in time. The loss has the same $\sin^2$ structure as the projection-misalignment loss at the snapshot level in [54], but it acts along the *temporal* direction (level versus velocity).

Setting. Let $\mathbf{z} = (D_t, V_t)^\top$ be the level and the velocity at inspection time t, with Gaussian prior $\mathcal{N}(\mathbf{z}_0, \boldsymbol{\Sigma})$, observed through $\mathbf{X}_{1:t}$ in a linear-Gaussian model (for example the local-linear-trend state space [62, 59] of Section 3.1 with Gaussian sensor noise). The posterior is $\mathcal{N}(\hat{\mathbf{z}}, \mathbf{P})$ where $\hat{\mathbf{z}} = \mathbb{E}[\mathbf{z} \mid \mathbf{X}_{1:t}]$ and $\mathbf{P}$ does not depend on the data (as for the covariance recursion of the Kalman filter [58, 59]); $\hat{\mathbf{z}} \sim \mathcal{N}(\mathbf{z}_0, \boldsymbol{\Lambda})$ with $\boldsymbol{\Lambda} = \boldsymbol{\Sigma} - \mathbf{P} \succ 0$ (law of total variance [57, Sect. 1.5.5]), and $\hat{\mathbf{z}}$ is a sufficient statistic of $\mathbf{X}_{1:t}$ for $\mathbf{z}$. The target is the frozen-velocity RUL $m(\mathbf{z}) = (D_{\text{fail}} - D_t)/V_t$, which is nonlinear in $\mathbf{z}$. We linearise it at a *fixed*, non-random operating point $\bar{\mathbf{z}} = (\bar{D}, \bar{V})^\top$, for example the prior mean at the inspection stage. Then

$$\mathbf{g} = \nabla m(\bar{\mathbf{z}}) = -\frac{1}{\bar{V}}\Big(1,\ \frac{\bar{a}}{\bar{V}}\Big)^\top, \qquad \bar{a} = D_{\text{fail}} - \bar{D}, \tag{16}$$

is a fixed vector, and $m_L(\mathbf{z}) = m(\bar{\mathbf{z}}) + \mathbf{g}^\top(\mathbf{z} - \bar{\mathbf{z}})$ is a fixed linear functional of $\mathbf{z}$. All statements of this section are exact for m_L; the error of replacing m by m_L is not covered by the theorems and is examined numerically in Section 5.1. A scalar *linear* HI built from the sufficient statistic, with fixed weights $\mathbf{u}$, is $h_{\mathbf{u}} = \mathbf{u}^\top\hat{\mathbf{z}}$. Define the signal-to-noise ratio $\rho = \mathbf{g}^\top\boldsymbol{\Lambda}\mathbf{g}/\mathbf{g}^\top\mathbf{P}\mathbf{g}$ and the $\boldsymbol{\Lambda}$-angle $\psi \in [0, \pi/2]$ between $\mathbf{g}$ and $\mathbf{u}$,

$$\cos\psi = \frac{|\mathbf{g}^\top\boldsymbol{\Lambda}\mathbf{u}|}{\sqrt{(\mathbf{g}^\top\boldsymbol{\Lambda}\mathbf{g})(\mathbf{u}^\top\boldsymbol{\Lambda}\mathbf{u})}}. \tag{17}$$

Lemma 1 (Reduction to functionals of $\hat{\mathbf{z}}$)**.** *For every scalar linear functional $w^\top\mathbf{X}_{1:t}$ there is $\mathbf{u}$ such that $\operatorname{corr}(m_L, w^\top\mathbf{X})^2 \le \operatorname{corr}(m_L, \mathbf{u}^\top\hat{\mathbf{z}})^2$. Hence no scalar linear HI is more informative about m_L than the best $h_{\mathbf{u}}$. Here* corr *denotes the correlation coefficient.*

Proof. Let $\mathbf{C} = \operatorname{Cov}(\mathbf{z}, \mathbf{X})$ and $\mathbf{V} = \operatorname{Var}(\mathbf{X})$, so $\hat{\mathbf{z}} = \mathbf{z}_0 + \mathbf{C}\mathbf{V}^{-1}(\mathbf{X} - \mathbb{E}\mathbf{X})$ and $\boldsymbol{\Lambda} = \mathbf{C}\mathbf{V}^{-1}\mathbf{C}^\top$. Put $\mathbf{u} = \boldsymbol{\Lambda}^{-1}\mathbf{C}w$. Then $\operatorname{Cov}(\hat{\mathbf{z}}, w^\top\mathbf{X}) = \mathbf{C}\mathbf{V}^{-1}\mathbf{V}w = \mathbf{C}w = \boldsymbol{\Lambda}\mathbf{u}$, so $w^\top\mathbf{X} = \mathbf{u}^\top\hat{\mathbf{z}} + e$ with e uncorrelated with $\hat{\mathbf{z}}$ and, since $\operatorname{Cov}(\mathbf{z}, w^\top\mathbf{X}) = \mathbf{C}w = \boldsymbol{\Lambda}\mathbf{u} = \operatorname{Cov}(\mathbf{z}, \mathbf{u}^\top\hat{\mathbf{z}})$, also uncorrelated with $\mathbf{z}$. Therefore $\operatorname{Cov}(m_L, w^\top\mathbf{X}) = \operatorname{Cov}(m_L, \mathbf{u}^\top\hat{\mathbf{z}}) = \mathbf{g}^\top\boldsymbol{\Lambda}\mathbf{u}$ while $\operatorname{Var}(w^\top\mathbf{X}) = \mathbf{u}^\top\boldsymbol{\Lambda}\mathbf{u} + \operatorname{Var}(e) \ge \operatorname{Var}(\mathbf{u}^\top\hat{\mathbf{z}})$, which gives the claim. □

Theorem 4 (Angular loss law)**.** *In the setting above, for every fixed $\mathbf{u} \neq 0$:*

(a) $\operatorname{Var}(m_L \mid h_{\mathbf{u}}) = \mathbf{g}^\top\mathbf{P}\mathbf{g} + (\mathbf{g}^\top\boldsymbol{\Lambda}\mathbf{g})\sin^2\psi$.

(b) For the linearised target, with $I(m_L; \mathbf{X}_{1:t}) = \frac{1}{2}\ln(1+\rho)$,

$$\frac{\mathcal{B}_{\text{info}}(h_{\mathbf{u}})}{\mathcal{B}_{\min}} = 1 + \rho\sin^2\psi, \qquad \eta_m(h_{\mathbf{u}}) = 1 - \frac{\ln(1+\rho\sin^2\psi)}{\ln(1+\rho)}. \tag{18}$$

(c) $\eta_m(h_{\mathbf{u}}) = 1$ if and only if $\mathbf{u} \parallel \mathbf{g}$; the HI $h_{\mathbf{g}} = \mathbf{g}^\top\hat{\mathbf{z}}$ is information-lossless for m_L and is, by Lemma 1, the best scalar linear HI of $\mathbf{X}_{1:t}$.

Proof. (a) Because $\mathbf{z} = \hat{\mathbf{z}} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \mathbf{P})$ independent of $\hat{\mathbf{z}}$, we have $\operatorname{Cov}(\mathbf{z}, \hat{\mathbf{z}}) = \boldsymbol{\Lambda}$ and $\operatorname{Var}(\mathbf{g}^\top\mathbf{z}) = \mathbf{g}^\top\boldsymbol{\Sigma}\mathbf{g} = \mathbf{g}^\top\boldsymbol{\Lambda}\mathbf{g} + \mathbf{g}^\top\mathbf{P}\mathbf{g}$. The pair $(\mathbf{g}^\top\mathbf{z}, h_{\mathbf{u}})$ is jointly Gaussian with $\operatorname{Cov} = \mathbf{g}^\top\boldsymbol{\Lambda}\mathbf{u}$ and $\operatorname{Var}(h_{\mathbf{u}}) = \mathbf{u}^\top\boldsymbol{\Lambda}\mathbf{u}$, so the Gaussian conditioning formula [57, Sect. 2.3] gives $\operatorname{Var}(m_L \mid h_{\mathbf{u}}) = \mathbf{g}^\top\boldsymbol{\Lambda}\mathbf{g} + \mathbf{g}^\top\mathbf{P}\mathbf{g} - (\mathbf{g}^\top\boldsymbol{\Lambda}\mathbf{u})^2/(\mathbf{u}^\top\boldsymbol{\Lambda}\mathbf{u}) = \mathbf{g}^\top\mathbf{P}\mathbf{g} + \mathbf{g}^\top\boldsymbol{\Lambda}\mathbf{g}\,(1 - \cos^2\psi)$.

(b) In the linearised model the conditional law of m_L is Gaussian, so the entropy power equals the variance. Taking the ratio of (a) to the case $h = \hat{\mathbf{z}}$, for which $\operatorname{Var}(m_L \mid \hat{\mathbf{z}}) = \mathbf{g}^\top\mathbf{P}\mathbf{g}$, gives the first formula. Moreover, by the formula for the mutual information of jointly Gaussian variables [40, Chap. 8], $I(m_L; \mathbf{X}) = \frac{1}{2}\ln(\operatorname{Var}(m_L)/\mathbf{g}^\top\mathbf{P}\mathbf{g}) = \frac{1}{2}\ln(1+\rho)$ and $I(m_L; h_{\mathbf{u}}) = \frac{1}{2}\ln\big((1+\rho)/(1+\rho\sin^2\psi)\big)$; their ratio is η_m.

(c) $\sin^2\psi = 0$ iff $\boldsymbol{\Lambda}^{1/2}\mathbf{g} \parallel \boldsymbol{\Lambda}^{1/2}\mathbf{u}$ iff $\mathbf{u} \parallel \mathbf{g}$ since $\boldsymbol{\Lambda} \succ 0$. For $\mathbf{u} = \mathbf{g}$ the loss vanishes, and Lemma 1 shows that no scalar linear functional of $\mathbf{X}$ can do better. □

Table 3. Inflation of the bound at $t = 50$, mean over 500 units: prediction at the fixed operating point, prediction at the unit's own estimate, and Monte Carlo for the exact target. ψ: angle to the unit-specific gradient.

HI	mean ψ [deg]	fixed point	unit gradient	Monte Carlo
level only $\mathbf{u} = (1, 0)$	6.0	1.179	1.177	1.181
velocity only $\mathbf{u} = (0, 1)$	5.8	1.164	1.194	1.197
stage-fixed gradient (prior mean)	0.7	1.000	1.004	1.003
unit-specific gradient	0.0	1.004	1.000	0.999

Remark 4 (Scope of the angular law)**.** *The theorem concerns scalar* linear *HIs with fixed weights and the linearised target. If the operating point is replaced by the posterior mean* $\hat{\mathbf{z}}$ *of each unit, the gradient becomes a function of the data, and the theorem then describes the loss conditionally on freezing the gradient, that is, a local analysis around the unit's own estimate. An HI with unit-specific weights is a nonlinear scalar function of* $\hat{\mathbf{z}}$ *and is not covered by the optimality statements, which also do not exclude nonlinear scalar HIs with fixed structure.*

Corollary 1 (No static scalar linear HI is optimal over the whole life)**.** *Let* $\mathbf{g}_1, \mathbf{g}_2$ *be the gradients at the operating points of two inspection stages with information matrices* $\mathbf{\Lambda}_1, \mathbf{\Lambda}_2$ *and ratios* $\rho_1, \rho_2 > 0$*. If* $\mathbf{g}_1 \nparallel \mathbf{g}_2$*, then for every fixed* $\mathbf{u}$ *the bound is inflated at one of the two stages,* $\max_j \rho_j \sin^2 \psi_j(\mathbf{u}) \geq \delta > 0$*, where* δ *depends only on* $(\mathbf{g}_j, \mathbf{\Lambda}_j, \rho_j)$*. If* $\mathbf{\Lambda}_1 = \mathbf{\Lambda}_2$*, then* $\delta \geq \min_j \rho_j \sin^2(\psi_{12}/2)$ *with* ψ_{12} *the* $\mathbf{\Lambda}$*-angle between* $\mathbf{g}_1$ *and* $\mathbf{g}_2$*. A stage-wise HI that uses the weights* $\mathbf{g}_j$ *at stage* j *has no inflation, relative to the linearised target, at any stage.*

Proof. By Theorem 4(c), inflation at stage j vanishes only for $\mathbf{u} \parallel \mathbf{g}_j$, so it vanishes at both stages only if $\mathbf{g}_1 \parallel \mathbf{g}_2$. The map $\mathbf{u} \mapsto \max_j \rho_j \sin^2 \psi_j(\mathbf{u})$ is continuous and positive on the unit circle, hence has a positive minimum δ. For the quantitative statement note that the $\mathbf{\Lambda}$-angle between lines satisfies the triangle inequality $\psi_{12} \leq \psi_1 + \psi_2$. Indeed, for unit vectors a, b in the $\mathbf{\Lambda}$-inner product the angle $\arccos\langle a, b\rangle$ is the geodesic distance on the unit sphere, which is a metric, and the angle between the lines through a and b is $\min_{s=\pm 1} \arccos\langle a, sb\rangle$. Choose the sign of the representative of $\mathbf{u}$ so that its angle to the representative of $\mathbf{g}_1$ is ψ_1, then the sign of the representative of $\mathbf{g}_2$ so that its angle to that of $\mathbf{u}$ is ψ_2; the spherical triangle inequality bounds the angle between the representatives of $\mathbf{g}_1$ and $\mathbf{g}_2$ by $\psi_1 + \psi_2$, and ψ_{12} is not larger. Therefore $\max_j \psi_j \geq \psi_{12}/2$ and $\sin^2$ is increasing on $[0, \pi/2]$. The last sentence is Theorem 4(c) applied at each stage. □

Corollary 2 (Rotation of the optimal weights)**.** *In the coordinates* (D_t, V_t) *the optimal direction is* $\mathbf{g}_t \propto (1, \bar{\tau}_t)^\top$ *with* $\bar{\tau}_t = \bar{a}_t / \bar{V}_t$ *the frozen-velocity RUL (in steps) at the operating point of stage* t*. The optimal scalar linear HI therefore weights the velocity estimate* $\bar{\tau}_t$ *times more than the level estimate; the weight moves from the velocity (early life,* $\bar{\tau}_t$ *large) to the level (end of life,* $\bar{\tau}_t \to 0$*), and the Euclidean angle between the optimal directions at RULs* τ_1, τ_2 *is* $|\arctan \tau_1 - \arctan \tau_2|$.

Corollary 2 explains why an indicator built from a level-like feature, for example the first principal component of cumulative-damage features, loses information early in life, and why a velocity-like feature loses it late in life. It is the temporal counterpart of the snapshot-level rotation of the signal direction. It also motivates the predictor of Section 6, which uses both the smoothed HI level and its windowed slope: this pair estimates the two-dimensional sufficient statistic $\hat{\mathbf{z}}$.

5.1 Numerical check

The model is the local linear trend of Section 3.1 with the raw sensor and $D_{\text{fail}} = 200$. The posterior of $\mathbf{z} = (D_t, V_t)$ is Gaussian, and $\mathbf{\Lambda}$ is strongly anisotropic (eigenvalues 0.004 and 204). For each of 500 units, the exact entropy power of the *nonlinear* target $m = (D_{\text{fail}} - D_t)/V_t$ given $h_{\mathbf{u}}$ is estimated from the exact conditional Gaussian of $\mathbf{z}$ (2000 draws, Vasicek estimator) and compared with $1 + \rho \sin^2 \psi$ computed in two ways: at a fixed operating point, the prior mean of the stage, as in Theorem 4; and at the unit's own posterior mean, a local analysis in the sense of Remark 4. As a sanity check, in the constant-drift model of Section 4.1 $\mathbf{\Lambda}$ is effectively of rank one and the predicted inflation is 1.000 for every $\mathbf{u}$ (Monte Carlo: 1.000 to 1.003), as the remark after Theorem 3 implies.

At $t = 50$ both predictions agree with the Monte Carlo value within about 3% for all HIs (Table 3). Figure 4 illustrates Corollary 1. The static HI aligned at $t^* = 50$ is lossless at $t = 25$ and 50 but inflates the bound by about 40% at $t = 75$ (Monte Carlo 1.39), the level-only HI by about 37% and the velocity-only HI by a factor of about 5. The prediction at the fixed prior-mean operating point agrees with Monte Carlo at $t = 25$ and 50 but underestimates the late-life inflation of the static and the velocity-only HI (1.12 and 3.5 at $t = 75$), because the gradient of the nonlinear target at the prior mean differs from that at the units' own estimates; the prediction with the

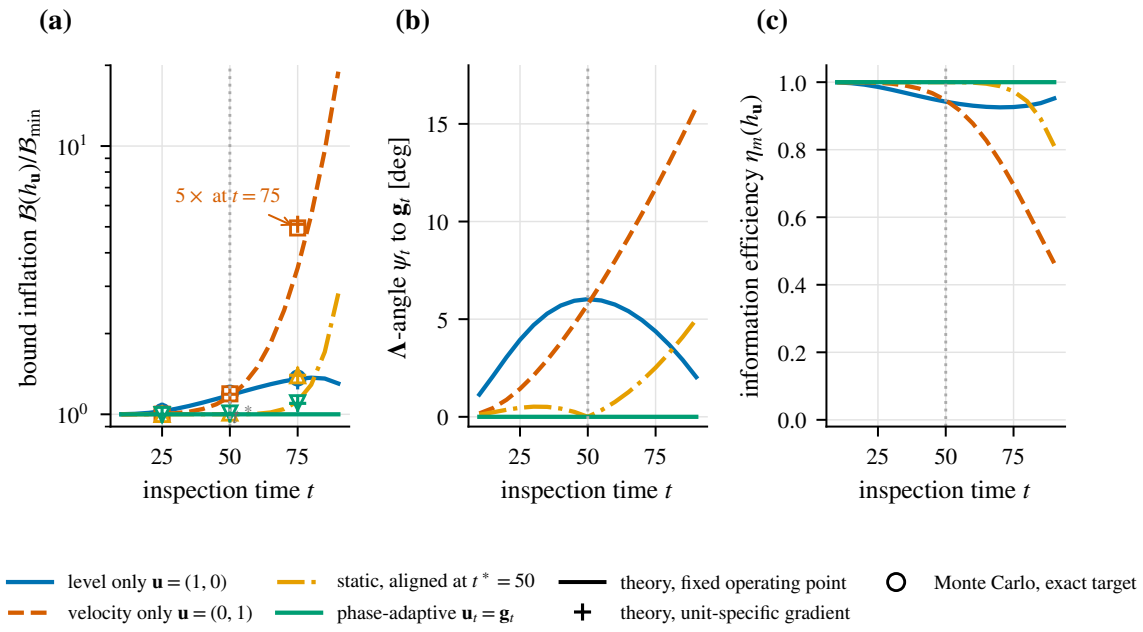


Figure 4. Angular loss law for scalar linear HIs. (a) Inflation of the bound. (b) Angle to the optimal direction $\mathbf{g}_t$ at the fixed operating point. (c) Information efficiency η_m. The static HI is aligned at $t^* = 50$; the phase-adaptive HI uses the stage-wise gradient at the prior mean.

unit's own gradient (1.40 and 5.1) agrees within 3%. The same limitation applies to the stage-wise HI: with weights fixed at the prior mean its inflation for the exact target is 1.10 at $t = 75$, whereas an HI with unit-specific weights stays within 1% of the minimum (1.005). The linear theory is therefore an accurate local approximation but not an exact statement for the nonlinear target.

5.2 Regime leakage

When the sensors are recorded under several operating regimes, the sensor responds to the degradation with a regime-dependent offset and gain, whereas the degradation process itself is the same. The following statement quantifies what a regime-blind read-out loses.

Proposition 5 (Regime leakage floor)**.** *Let a latent level D with mean $\bar{d}$ and variance v be observed through a sensor as $x = a_r D + b_r + \varepsilon$, where the regime r takes R values with probabilities π_r independently of D and of the noise ε (variance σ^2); a_r and b_r are the gain and offset of regime r. Put $\bar{a} = \mathbb{E}a_r$, $V_x = \mathbb{E}[a_r^2]\mathbb{E}[D^2] - (\bar{a}\bar{d})^2 + \mathrm{Var}(b_r) + 2\bar{d}\,\mathrm{Cov}(a_r, b_r) + \sigma^2$. The best regime-blind linear read-out of D from x has mean squared error*

$$\mathrm{MSE}_{\mathrm{blind}} = v - \frac{\bar{a}^2 v^2}{V_x}, \qquad \textit{and} \qquad \mathrm{MSE}_{\mathrm{blind}} \xrightarrow[\sigma \to 0]{} \frac{v\,[\mathrm{Var}(a_r)\,v + \mathrm{Var}(\bar{d}a_r + b_r)]}{\bar{a}^2 v + \mathrm{Var}(a_r)\,v + \mathrm{Var}(\bar{d}a_r + b_r)} > 0 \tag{19}$$

unless the regimes agree in a_r and in $\bar{d}a_r + b_r$, whereas a read-out that knows the regime has $\mathrm{MSE}_{\mathrm{aware}} = \sigma^2 \sum_r \pi_r / a_r^2 \to 0$.

Proof. Because D and r are independent, $\mathrm{Cov}(D, x) = \bar{a}\,v$, and $\mathrm{Var}(x) = \mathrm{Var}(a_r D) + \mathrm{Var}(b_r) + 2\bar{d}\,\mathrm{Cov}(a_r, b_r) + \sigma^2 = V_x$, since $\mathrm{Cov}(a_r D, b_r) = \bar{d}\,\mathrm{Cov}(a_r, b_r)$ and $\mathrm{Var}(a_r D) = \mathbb{E}[a_r^2]\mathbb{E}[D^2] - (\bar{a}\bar{d})^2$; the best linear predictor of D from x has error $v - \mathrm{Cov}(D, x)^2 / \mathrm{Var}(x)$ (the conditional variance of a Gaussian pair with the same second moments [57, Sect. 2.3]). For the limit use $\mathbb{E}[a_r^2]\mathbb{E}[D^2] - (\bar{a}\bar{d})^2 = \bar{a}^2 v + \mathrm{Var}(a_r)(v + \bar{d}^2)$ and $\mathrm{Var}(a_r)\bar{d}^2 + \mathrm{Var}(b_r) + 2\bar{d}\mathrm{Cov}(a_r, b_r) = \mathrm{Var}(\bar{d}a_r + b_r)$, so that at $\sigma = 0$, $V_x = \bar{a}^2 v + \mathrm{Var}(a_r)v + \mathrm{Var}(\bar{d}a_r + b_r)$. Given the regime, the read-out $(x - b_r)/a_r$ is unbiased with variance σ^2 / a_r^2. □

Normalising the sensors within each regime removes the regime offsets but not the regime-dependent gains, so the floor remains. In a numerical example with six regimes ($a_r \in [0.6, 1.6]$, $b_r \in [-0.5, 0.6]$, $v = 0.08$), the formula and a simulation agree on $\mathrm{MSE}_{\mathrm{blind}} = 0.060$ at $\sigma = 0$, so that only 25% of $\mathrm{Var}(D)$ is explained, whereas the regime-aware read-out has an error that tends to zero (Figure 5).

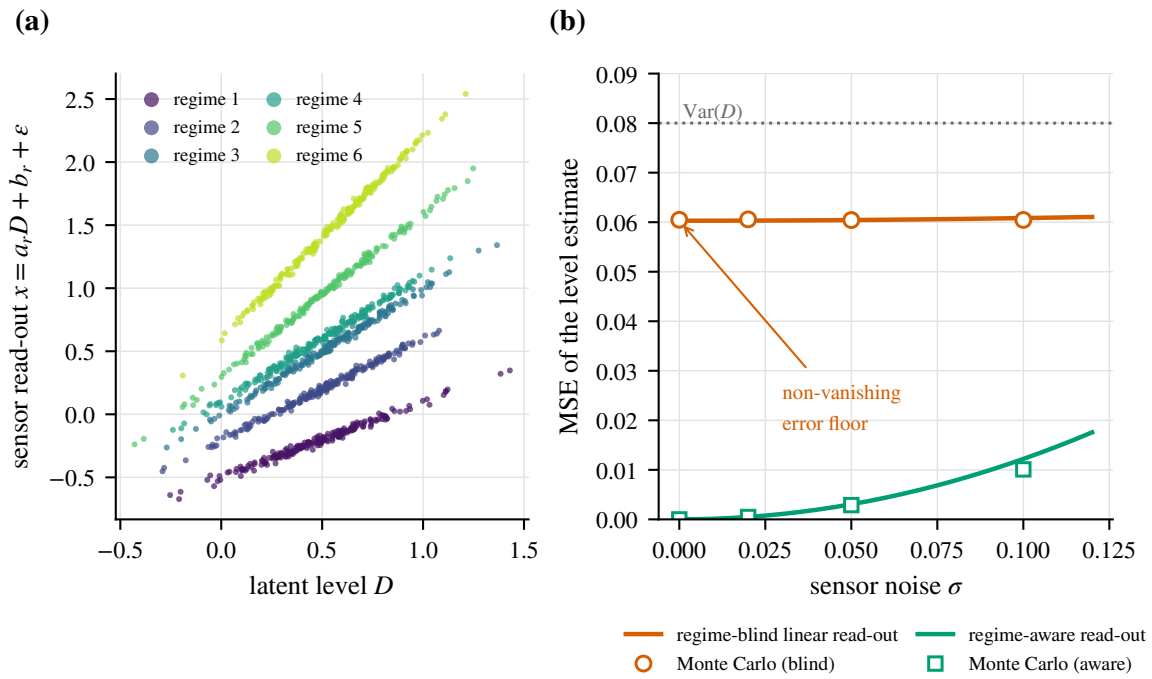


Figure 5. Regime leakage in the six-regime example. (a) Sensor read-out against latent level for each regime. (b) Error of the regime-blind and regime-aware read-outs: formula (lines) and simulation (markers).

6 Consequences for estimators

The results above concern information budgets and the structure of scalar indicators. This section derives what they imply for RUL estimators and describes the estimators used in the empirical assessment. The path-space objective and the error-accumulation result follow from the theory; the architectures and the calibration are design choices.

6.1 Path-space objective

We seek a causal stochastic encoder, the *physics-guided information bottleneck* (PGIB), $q_\theta(H_{1:t} \mid \mathbf{X}_{1:t}) = \prod_{k=1}^{t} q_\theta(H_k \mid H_{k-1}, \mathbf{X}_{1:k})$, whose increments are Gaussian, $\mathcal{N}\big(H_{k-1} + u_\theta(\mathbf{X}_{1:k})\Delta t,\ s_\theta^2(\mathbf{X}_{1:k})\Delta t\big)$, with a positive velocity u_θ (enforced by a softplus output) and a variance s_θ^2. The encoder is regularised towards a physics prior $p_\vartheta(H_{1:t}) = \prod_k \mathcal{N}\big(H_{k-1} + \eta(H_{k-1};\vartheta)\Delta t,\ \sigma_B^2 \Delta t\big)$, $H_0 \equiv 0$, where Δt is the time step, η a state-dependent drift with parameters ϑ, and σ_B the diffusion coefficient.

Proposition 6 (Compression bound and path-space KL)**.** *For any prior p_ϑ, $I(\mathbf{X}_{1:t}; H_{1:t}) \le \mathbb{E}_{\mathbf{X}}\big[D_{\mathrm{KL}}(q_\theta(H_{1:t} \mid \mathbf{X}_{1:t}) \,\|\, p_\vartheta(H_{1:t}))\big]$, and for the Gaussian factorisation above the divergence is exactly*

$$D_{\mathrm{KL}} = \frac{1}{2} \sum_{k=1}^{t} \mathbb{E}_{q_\theta(H_{k-1} \mid \mathbf{X}_{1:k-1})} \left[\frac{s_\theta^2}{\sigma_B^2} - 1 - \ln \frac{s_\theta^2}{\sigma_B^2} + \frac{(u_\theta - \eta(H_{k-1};\vartheta))^2 \, \Delta t}{\sigma_B^2} \right]. \tag{20}$$

Proof. The first claim is the standard variational upper bound of the information bottleneck [41, 42, 43]: $I(\mathbf{X}; H) = \mathbb{E}_{\mathbf{X}} D_{\mathrm{KL}}(q(H \mid \mathbf{X}) \| q(H)) \le \mathbb{E}_{\mathbf{X}} D_{\mathrm{KL}}(q(H \mid \mathbf{X}) \| p(H))$ because the difference is $D_{\mathrm{KL}}(q(H) \| p(H)) \ge 0$. The chain rule of the KL divergence [40, Sect. 2.5] over the causal factorisation gives $\sum_k \mathbb{E}_{q(H_{k-1} \mid \cdot)} D_{\mathrm{KL}}(q(H_k \mid H_{k-1}, \mathbf{X}_{1:k}) \| p_\vartheta(H_k \mid H_{k-1}))$, and the Gaussian KL with means differing by $(u_\theta - \eta)\Delta t$ and variances $s_\theta^2 \Delta t$, $\sigma_B^2 \Delta t$ yields the summand (the Gaussian KL of the variational autoencoder [20]). □

The training objective is $\mathcal{L}_{\mathrm{PGIB}} = \mathcal{L}_{\mathrm{FHT}} + \beta\, D_{\mathrm{KL}}$, where β weights the compression term and $\mathcal{L}_{\mathrm{FHT}}$ is the negative log-likelihood of the observed failure time under the predictive law below, a variational surrogate for $-I(T_t; H_{1:t})$ in the sense of [42] (Algorithm 1 in Appendix A).

6.2 Predictive law

Let the posterior of the unit's drift μ given the HI increments be $\mathcal{N}(\hat{\mu}_t, \sigma_{\mu,t}^2)$ (conjugate update). Marginalising the inverse-Gaussian law over μ (a Gaussian integral; random-drift Wiener models

are treated in [28, 29]) gives

$$p(T_t = \tau \mid H_{1:t}) = \frac{a}{\sqrt{2\pi\tau^3(\sigma_B^2 + \sigma_{\mu,t}^2\tau)}} \exp\left(-\frac{(a - \hat{\mu}_t\tau)^2}{2\tau(\sigma_B^2 + \sigma_{\mu,t}^2\tau)}\right), \qquad a = D_{\text{fail}} - \hat{D}_t. \tag{21}$$

Here σ_B^2 is the aleatoric and $\sigma_{\mu,t}^2\tau$ the epistemic contribution to the effective diffusion; as t grows $\sigma_{\mu,t}^2 \to 0$ and (21) reduces to the inverse-Gaussian law. It is a defective density, the missing mass being the probability of a non-positive drift, and $\hat{D}_t$ is the filtered state rather than the raw HI value, which avoids the degeneracies of non-monotone paths. The law was checked against numerical integration of the mixture (Section 4.1).

6.3 Observation-corrected and regime-aware encoders

The encoder above integrates a positive velocity, $H_k = H_{k-1} + u_\theta\Delta t + \ldots$, so nothing anchors the level to the observations. Section 5 treats the level and the velocity as latent quantities, and the level must be re-estimated from the sensors at every step; this is the measurement update of a Kalman filter [58]. The observation-corrected encoder, PGIB-KF (KF: Kalman filter), therefore adds a level read-out $l_k = f_l(\mathbf{h}_k)$ and a learned gain $\omega_k = \sigma(f_\omega(\mathbf{h}_k)) \in (0, 1)$ to the encoder features $\mathbf{h}_k$,

$$H_k = (1 - \omega_k)\big(H_{k-1} + u_k + s_k\epsilon_k\big) + \omega_k\, l_k. \tag{22}$$

Given H_{k-1} this is again a Gaussian transition with drift $u_k^{\text{eff}} = (1 - \omega_k)u_k + \omega_k(l_k - H_{k-1})$ and standard deviation $s_k^{\text{eff}} = (1 - \omega_k)s_k$, so Proposition 6 applies unchanged: the bottleneck now prices the use of the observation beyond what the physics prior allows ($\beta \to \infty$ gives $\omega \to 0$, a pure integrator). The HI is no longer monotone by construction.

Proposition 7 (Error accumulation without observation correction)**.** *Let the velocity estimate carry the error $e_j = b + w_j$ with a unit-specific persistent bias b and white noise w_j of variance σ_w^2, and let the level read-out carry independent zero-mean noise of variance σ_n^2. For the pure integrator ($\omega = 0$) the level error after t steps has mean tb and variance $t\sigma_w^2$, both unbounded in t. For a constant gain $\omega \in (0, 1]$ the level error converges to a stationary law with*

$$\mathbb{E}\,\varepsilon_\infty = \frac{(1 - \omega)\,b}{\omega}, \qquad \operatorname{Var}\varepsilon_\infty = \frac{(1 - \omega)^2\sigma_w^2 + \omega^2\sigma_n^2}{\omega(2 - \omega)}. \tag{23}$$

Proof. The level error obeys $\varepsilon_k = (1 - \omega)(\varepsilon_{k-1} + e_k) + \omega n_k$ with $\varepsilon_0 = 0$; write $\alpha = 1 - \omega$. For $\omega = 0$, $\varepsilon_t = \sum_{j \le t} e_j$ has mean tb and variance $t\sigma_w^2$. For $\omega > 0$ the mean satisfies $m_k = \alpha(m_{k-1} + b)$, with fixed point $\alpha b/(1 - \alpha) = (1 - \omega)b/\omega$ approached geometrically, and the variance $v_k = \alpha^2 v_{k-1} + \alpha^2\sigma_w^2 + \omega^2\sigma_n^2$ (independent noise terms) has fixed point $(\alpha^2\sigma_w^2 + \omega^2\sigma_n^2)/(1 - \alpha^2)$ with $1 - \alpha^2 = \omega(2 - \omega)$. □

A simulation ($b = 0.01$, $\sigma_w = 0.02$, $\sigma_n = 0.15$, 400 steps, 2×10^5 paths) agrees with both expressions (for $\omega = 0$ the bias is $4.00 = tb$; for $\omega = 0.2$ the bias and variance are 0.0399 and 0.00321 against 0.04 and 0.00321). A constant gain amounts to exponential smoothing, the steady state of the local-level filter [62, 59]. A persistent velocity bias is the generic situation when units differ in initial offset or wear, and it is the mechanism that we diagnose for the integrating encoder on the turbofan data (Section 7).

By Proposition 5, a regime-blind read-out has an error floor, and per-regime normalisation removes the offsets but not the gains. The regime-aware variant, *PGIB-KFR*, therefore places the regime in the *observation* model, where the proposition locates it. The process model is shared, whereas the read-out l_k and the gain ω_k use regime-specific linear heads on the encoder features, $l_k = \mathbf{h}_k^\top \mathbf{W}_{r_k}^l + b_{r_k}^l$ and $\omega_k = \sigma(\mathbf{h}_k^\top \mathbf{W}_{r_k}^\omega + b_{r_k}^\omega)$, where r_k is the regime, which is also an input of the encoder. The regimes are the clusters of the standardised operating settings. With a single regime PGIB-KFR reduces to PGIB-KF, which we check on the single-regime subsets.

6.4 Calibrated predictive law with age fusion

Because the age is always known (Proposition 2), the HI-based law $p_{\text{HI}}(\tau)$, which is (21) with diffusion $c\,v_t$ and drift variance $c\,v_t/w_t$, is pooled log-linearly [63, 64] with an age law. Here v_t is the variance of the HI increments over the trailing window of length w_t for non-learned HIs and the learned σ_B^2 for the bottleneck models, and $c > 0$ is a calibration multiplier. The age law is $p_{\text{age}}(\tau \mid t) = f_L(t + \tau)/S_L(t)$, the log-normal lifetime density [65] fitted on training lifetimes (log-scale floor 0.15) conditioned on survival to age t. The pooled law is

$$p(\tau) \;\propto\; p_{\text{HI}}(\tau)^{w_{\text{h}}}\, p_{\text{age}}(\tau \mid t)^{w_{\text{a}}}, \qquad (c, w_{\text{h}}, w_{\text{a}}) \in \mathcal{G}, \tag{24}$$

on a finite grid $\mathcal{G}$ ($c \in [10^{-2}, 10^{2}]$ on a log scale, ten pairs $(w_{\mathrm{h}}, w_{\mathrm{a}})$ including $(1, 0)$ and $(0, 1)$). The parameters minimise the negative log-likelihood of the true RUL, a strictly proper scoring rule [66], on HI paths produced *out of sample*: each HI construction is fitted on all but one training unit and applied to the left-out unit; the test unit is never used. The family contains the plain HI law and the age-only law, so in-sample the pooled law cannot do worse than either. Weights below one temper the corresponding law and widen the predictive interval, which the otherwise under-covering plain law requires. The pooling is a calibration device, not a Bayes-exact fusion, which would also require conditional independence of the HI and the age given the RUL, and division by the marginal law of the RUL. The same predictor is used for every HI, and a variant without fusion ($w_{\mathrm{a}} = 0$) is reported for each HI, so that the effects of the HI and of the fusion can be separated.

7 Empirical assessment

The theory makes four predictions that real run-to-failure data can test. (i) The baseline for an HI is the RUL given the age, so a sensor-free predictor that uses only the age should be competitive when the sensors are weak (Proposition 2). (ii) An integrating encoder accumulates error and an observation-corrected one does not (Proposition 7). (iii) Learned encoders have a training variance that must be measured before differences between HIs are interpreted. (iv) The error of a predictor certifies a bound on the information that its HI retains, which is vacuous when the error is large (Proposition 1).

Design. The data are battery cells from the NASA Prognostics Center of Excellence (PCoE) and from the Center for Advanced Life Cycle Engineering (CALCE, CS2 cells, in sets of four and six); bearings from the FEMTO-ST institute (PRONOSTIA, conditions 1 to 3) and from the NASA Center for Intelligent Maintenance Systems (IMS); and the four subsets FD001 to FD004 of the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) turbofan data, of which FD001 and FD003 have one operating regime and FD002 and FD004 six. Three simple HIs are compared with three learned encoders. The simple HIs are the first principal component (PCA-HI), a linear combination optimised for the sum of the geometric criteria monotonicity, trendability and prognosability (Geo-opt HI), and a supervised ridge regression (ridge HI). The learned encoders are a β-variational autoencoder (β-VAE), a variational information bottleneck without physics prior (VIB), and the PGIB of Section 6 with its Kalman-corrected (PGIB-KF) and regime-aware (PGIB-KFR) variants. All HIs feed the *same* calibrated predictor of Section 6.4. Evaluation is leave-one-unit-out for the small sets and uses the official test engines for C-MAPSS. Data, protocols, their history and the evidence tiers (which results are clean confirmations and which are development results) are given in Appendix A, and full tables in Appendix B.

Age and few training units (prediction i). With two to six training units, the age law has the lowest RMSE on both FEMTO conditions, a simple HI on three sets, the PGIB on one (four-cell CALCE) and the β-VAE on one (FEMTO condition 3). The age law is within a factor of 1.5 of the best method on six of the seven sets (Figure 6). On C-MAPSS the sensors are far more informative: the age law has an RMSE of 36.6 to 54.7 cycles, against 21.8 to 29.7 for the best simple HI. The age law is thus a strong baseline where the sensors are weak, and it must be reported.

Error accumulation and its repair (prediction ii). With 80 to 208 training engines, the integrating encoder is well behind the simple HIs and no better than the age law on FD001 to FD003 (single-model RMSE 34.8 to 48.7). The observation-corrected PGIB-KF reduces the RMSE to 23.8–40.7 for a single model and to 21.3–35.8 for a ten-seed ensemble (Table 4). The deficit is therefore not caused by the bottleneck objective, which is the same in both models, but by the missing anchoring of the level.

Training variance (prediction iii). A single learned encoder varies from seed to seed by 2.2 to 7.6 RMSE units, as much as the differences between competing HIs (Figure 7), which calls for variance-aware comparisons [67]. A three-seed ensemble on FD003 suggested that PGIB-KF beats the best simple HI (19.4 against 21.9), but with ten seeds [38] the difference disappears (23.3; paired difference $+1.4$, interval -1.6 to 4.5). With ten seeds, the ensembles of the observation-corrected encoders are statistically tied with the best simple or supervised HI on all four subsets, except PGIB-KF on FD004, where it is worse, and they are never significantly better. The regime-aware variant is not shown to help: it is worse than PGIB-KF with a single regime and better only on FD004, where the difference is not significant.

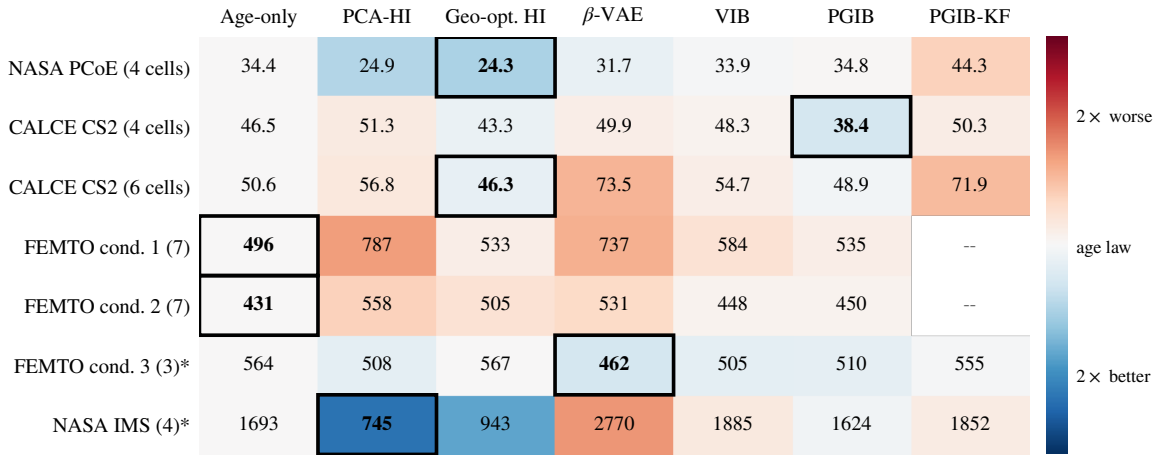


Figure 6. RMSE of the compared HIs on the seven small data sets (cycles for batteries, 10 s snapshots for bearings). Colour: ratio to the age law; frame: lowest value of each row; *: sets not used in the design.

Table 4. C-MAPSS, ten seeds per learned encoder. Single: RMSE, mean ± standard deviation over seeds; ens. 10: ten-seed ensemble; last column: paired difference of PGIB-KF from the best simple HI with 95% bootstrap interval over engines [68].

Subset	Best simple HI (RMSE)	PGIB-HI single	PGIB-KF-HI single	PGIB-KFR-HI single	PGIB-KF-HI ens. 10	PGIB-KFR-HI ens. 10	KF − best paired, ens. 10 [95% CI]	KFR − best
FD001	PCA (21.8)	34.8±2.9	23.8±3.3	27.5±3.2	21.3	25.5	-0.6 [-3.2, +2.5]	+3.6 [+0.2, +7.0]
FD002	regime ridge (28.4)	36.3±1.5	31.0±2.2	30.6±2.2	29.2	29.1	+0.8 [-0.7, +2.3]	+0.7 [-1.2, +2.8]
FD003	Geo-opt. (21.9)	41.2±1.9	30.1±4.6	33.8±6.5	23.3	28.4	+1.4 [-1.6, +4.5]	+6.5 [+2.4, +10.6]
FD004	Geo-opt. (29.7)	48.7±2.5	40.7±7.6	36.8±4.0	35.8	31.8	+6.1 [+2.5, +9.7]	+2.1 [-0.9, +5.0]

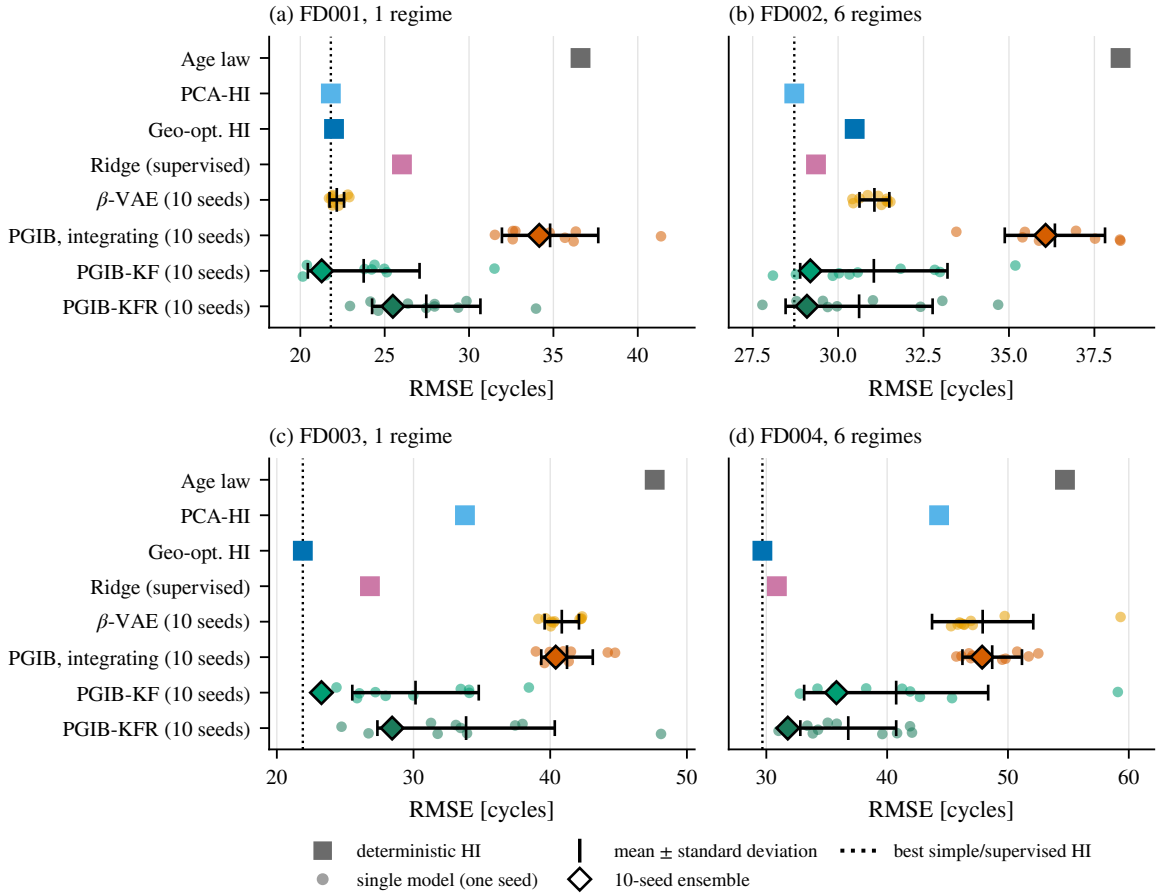


Figure 7. C-MAPSS test RMSE. Squares: deterministic HIs; dots: single models of each learned encoder; bars: mean and standard deviation; diamonds: ten-seed ensembles; dotted line: best simple or supervised HI.

Role of the age fusion. The age fusion lowers the RMSE of most HIs considerably (for example PCA-HI on CALCE with four cells from 186 to 51 cycles) but not always (NASA), and it narrows

the differences between HIs. Without fusion, that is, with a predictor based on the HI alone, the PGIB has the lowest RMSE on CALCE (four and six cells) and FEMTO condition 2, VIB on FEMTO condition 1, the β-VAE on FEMTO condition 3 and PCA-HI on NASA and IMS (Table 13). The conclusion that no learned encoder is superior therefore holds for the common age-fused predictor and is more mixed for HI-only predictors, for which the information in the HI is less masked by the age prior.

Certificates (prediction iv). The certified information is vacuous for NASA, the FEMTO conditions and, with one exception, IMS. It is positive only on CALCE, where the PGIB certifies 0.60 nats against 0.41 for the age law with four cells (Appendix B). These values are lower bounds on $I(T_t; H_{1:t})$, which contains the age, and on $I(T_t; t)$, respectively; their difference is not a lower bound on the information that the HI adds to the age. An age-conditional certificate (Proposition 2) requires $h(T_t \mid t)$; under a fitted lifetime model the age-only predictor itself obtains a positive value on four of the seven small sets, so these data do not allow us to separate the information of the HI from that of the age (Appendix B). With 77 or more units the age-conditional certificate passes this control and is informative (Section 8).

8 Real-data tests of the theory

Section 7 evaluates estimators. This section tests the predictions of Sections 2 to 5 directly, on two domains with many units: turbofan engines (the four C-MAPSS subsets, 100 to 260 training engines) and lithium-ion cells (HUST, 77 cells; MATR, 169 cells). All analyses use unit-grouped five-fold cross-validation: preprocessing, HIs and predictors are fitted on the training units of a fold and evaluated on its held-out units. The population consists of pairs of a unit and an age t between a minimum age (30 cycles for engines, 100 and 50 cycles for the HUST and MATR cells) and the age at which 10% of the training units are still alive. Pooling all ages of all units gives $T_t \mid t$ the law of $L - t$ given $L > t$ for the lifetime L, so that $h(T_t \mid t)$ is estimated from the lifetime distribution of the training units (Gaussian kernel density of $\ln L$) without further assumption. Predictors are gradient-boosted trees on summaries of the HI trajectory and the age, and confidence intervals are bootstrap intervals over units (Appendix D).

8.1 Turbofan engines

Age-conditional certificate (Proposition 2). The certificate $\bar{h}(T_t \mid t) - \frac{1}{2}\ln(2\pi e\,\mathrm{MSE})$ is a lower bound on $I(T_t; H_{1:t} \mid t)$. A predictor that uses only the age is a negative control, since it cannot exceed the bound; its certificate is -0.11, -0.11, -0.22 and -0.12 nats on FD001 to FD004. The HI trajectory certifies information beyond the age (Figure 8(a)): the ridge HI certifies 0.31, 0.21, 0.10 and 0.18 nats, with 95% intervals excluding zero except on FD003. Level-only and slope-only summaries of the same HI certify less than the full trajectory on FD001, FD002 and FD004 (0.07 to 0.17 nats on FD001), and a predictor on the raw sensors certifies 0.16 to 0.25 nats, similar to the supervised HI (and higher on FD003 and FD004), so on these data the HI is a compression of the sensors that retains the certified information at best, not an addition to it. With 100 or more engines the certificate is therefore non-vacuous and passes its control.

Squeezing (Theorem 2). A squeeze toward the mean HI trajectory of the training engines, with noise chosen as in Section 3.1, raises the monotonicity of the ridge HI from 0.13–0.24 to 0.41–0.58, where it saturates for $\gamma \geq 0.97$ at the value predicted from the template (0.55, 0.58, 0.42, 0.41 observed against 0.55, 0.61, 0.41, 0.41 predicted for FD001 to FD004; Figure 8(b)). Over the same range the certified information falls from 0.10–0.31 nats to about zero or below (Figure 8(c)). Monotonicity and information are decoupled in real engine data as in the simulation.

Direction of a scalar linear HI (Theorem 4, Corollary 1). For the level and the velocity of the ridge HI, the direction of the best linear scalar combination of the RUL moves across stages (for example between 21° and 56° across age stages on FD002, Table 21), and level-only and velocity-only HIs lose up to 20% and 38% of RMSE relative to the free direction (Figure 9(b,c)). However, the best *static* direction, combined with a stage-specific affine map, loses at most 4% in every stage and subset. The law of Section 5 is therefore consistent with the observed rotation, but its consequence is small on these data: it is material only when the angle between the stage optima is large relative to the signal-to-noise ratio, which is not the case here.

Regime leakage (Proposition 5). For a linear read-out of the life fraction, regime blindness increases the RMSE from 0.137 to 0.148 on FD002 and from 0.165 to 0.199 on FD004 (8% and 21%); with one regime the three read-outs coincide (Table 22). Per-regime normalisation removes most of the loss, and regime-specific weights add only 1%, so the residual effect of heterogeneous gains is small in these data.

Certificates of the benchmark methods. Using the same lifetime law, the certificates of the methods of Section 7 on the official test engines (Figure 9(a), Table 23) provide an information-theoretic reading of the error accumulation of Proposition 7. On FD001, FD002 and FD004 the age law passes the control (0.07, 0.01, 0.05 nats). The integrating PGIB certifies only 0.14, 0.07 and 0.19 nats, which is close to the age law on FD001 and FD002, whereas the observation-corrected PGIB-KF certifies 0.61, 0.28 and 0.48 nats and the best simple HI 0.59, 0.29 and 0.66 nats. On FD003 the control fails (age law: 0.33 nats, interval 0.23 to 0.44), so the lifetime law of the training engines does not describe the test population there and the FD003 values are not interpreted.

8.2 Lithium-ion cells

Data and indicators. The HUST collection [69] contains 77 lithium iron phosphate/graphite cells, each cycled under its own multi-stage charge and discharge protocol, with end of life at 80% of the initial capacity (977 to 2334 cycles; coefficient of variation of the lifetime 0.19). The MATR collection [22, 70] contains 169 cells of the same chemistry cycled under fast-charging protocols, with end of life at 80% of the nominal capacity (147 to 2235 cycles; coefficient of variation 0.42). Both were taken from the processed BatteryLife repository [71]. The HIs use only features of the charging phase (charge duration and charge voltage at 0.3 and 0.8 Ah), each expressed relative to cycles 2 to 11 of the same cell. The discharge capacity, which defines the end of life, enters only a reference predictor. Time scales are matched to the lifetimes (Appendix D).

Certificate and squeeze. The age-only control is at or below zero (-0.02 nats on HUST and -0.27 on MATR), and the ridge HI certifies 0.30 and 0.32 nats with intervals above zero (Figure 8(a), Table 19). The three charging features used directly certify more (0.67 and 0.61 nats) than the scalar HI constructed from them, as the data-processing inequality requires, and the capacity reference does not exceed them (0.59 and 0.29 nats). Squeezing raises the monotonicity of the ridge HI from 0.32 to 0.81 on HUST and from 0.39 to 0.48 on MATR, at the predicted limits (0.82 and 0.49), while the certificate falls from 0.30 to 0.00 and from 0.32 to -0.20 nats (Figure 8(b,c)). As for engines, monotonicity and information are decoupled.

Direction of a scalar linear HI. On HUST the optimal direction moves from 71°–80° in the first three life stages to 3°–16° in the last two, and the best static direction loses up to 9% of RMSE in a stage (18% in the last age stage), against up to 54% for velocity alone (Figure 9(b,c), Table 21). On MATR the optimal direction also rotates across the life fraction (88° to 42°), but a static direction loses at most 1.2%. The loss predicted by the angular law is thus material in one of the three data families with many units and negligible in the others.

Protocol leakage. A linear read-out of the life fraction from the raw charging features, standardised over the population, has an RMSE of 0.136 on HUST and 0.179 on MATR; expressing each feature relative to the own early cycles of the cell, which removes the protocol- and cell-specific level, lowers it to 0.115 and 0.142 (18% and 26%; Table 22). This is the analogue of regime normalisation of Proposition 5, with the cell playing the role of the regime.

9 Discussion and conclusions

Theory. The paper gives bounds, identities and laws for the information that an HI can retain about the RUL. The entropy-power bound lies below the Bayes-optimal error in every simulated case, and the gap equals the weighted non-Gaussianity of the posterior. A squeeze loses state information while monotonicity saturates, so geometric scores do not measure what matters. The error splits into an aleatoric floor, independent of the HI, and an epistemic term that grows with the information lost. A static scalar linear HI inflates the bound by $1 + \rho \sin^2 \psi$, which agrees with Monte Carlo within about three per cent when the linearisation is made at each unit's own estimate, so that no static scalar linear HI is optimal over the whole life; a regime-blind linear read-out keeps an error floor. Tests on turbofan engines (100 to 260 units) and lithium-ion cells (77

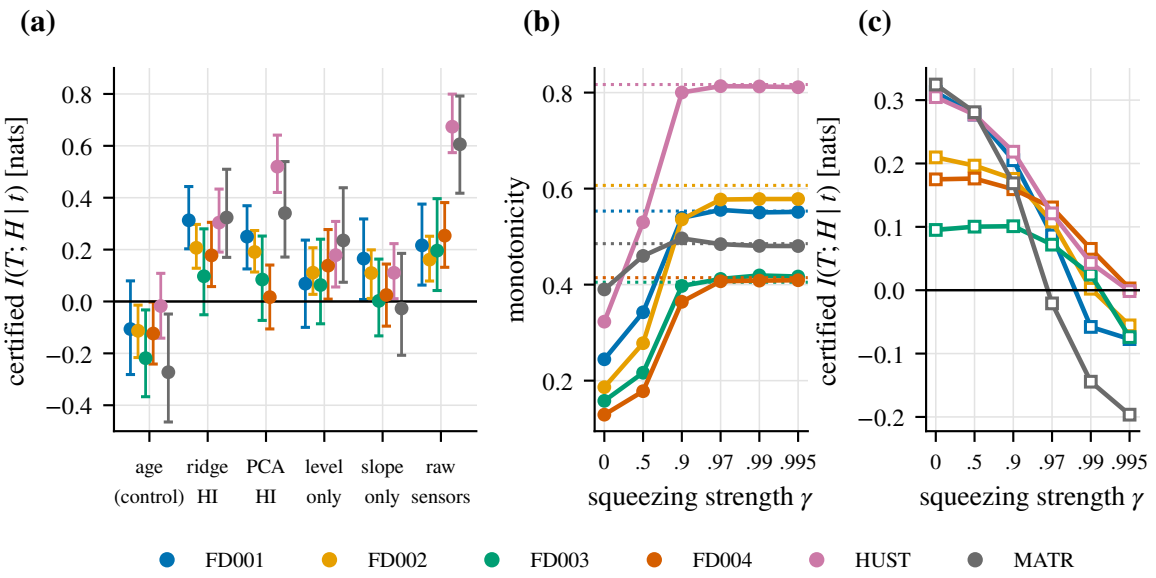


Figure 8. Cross-validated tests on turbofan engines (FD001–FD004) and lithium-ion cells (HUST, MATR). (a) Certified $I(T_t; H_{1:t} \mid t)$ of HI summaries, with 95% bootstrap intervals. (b) Monotonicity and (c) certified information after a squeeze of strength γ; dotted lines: predicted Mon_∞.

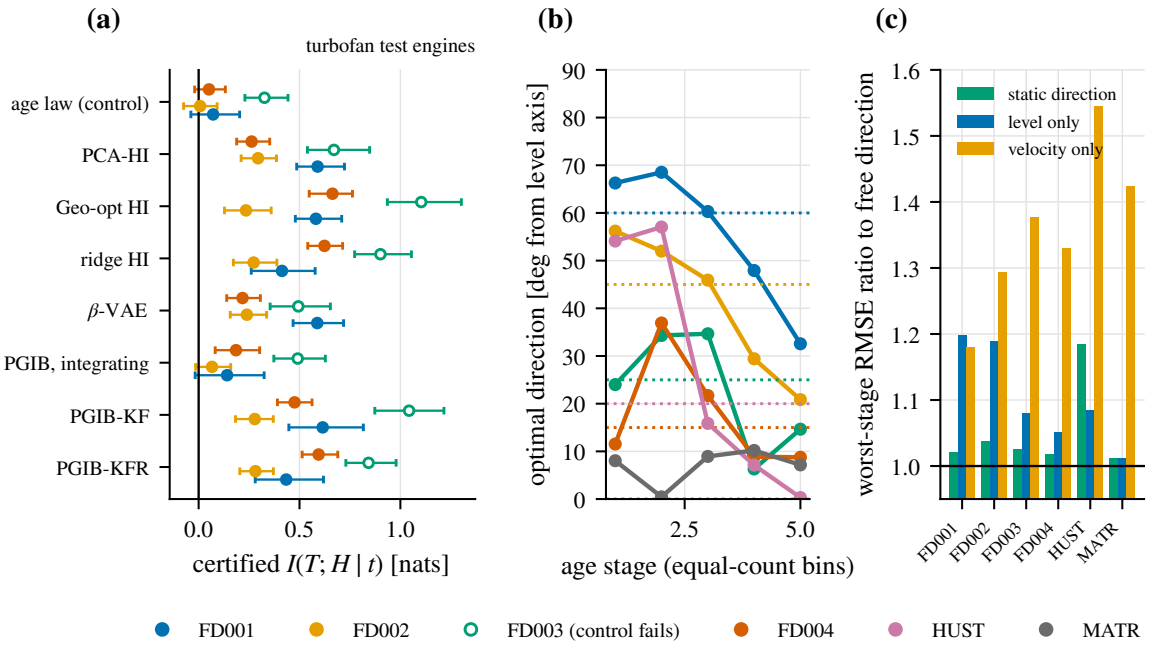


Figure 9. (a) Certificates of the benchmark methods on the official turbofan test engines (hollow markers: age-only control fails). (b) Optimal direction of a scalar linear HI per age stage; dotted: best static direction. (c) Worst-stage RMSE ratio to the free direction.

and 169 units) support the age-conditional certificate and the decoupling of monotonicity from information. They show that the observation-corrected encoder retains information that the integrating encoder loses, and that the consequence of the angular law is small for engines and for one cell collection but reaches 9 to 18% of the RMSE for the other. These statements concern budgets and structure; they do not say that a particular encoder reaches the budget, nor do they rank encoders.

Design implications and empirical findings. The theory suggests four design rules: keep the age in the baseline and the predictor (Proposition 2); complement a scalar HI with estimates of level and velocity (Corollary 2), as the windowed slope does in our predictor; anchor the level of an encoder to the observations (Proposition 7); and let the observation model depend on the regime (Proposition 5). The experiments confirm the first and the third. They neither confirm nor refute

the fourth, because the effect of the regime-aware encoder is within the training variability. No learned encoder is superior to simple indicators: with two to six training units a simple indicator or the age law is best on five of seven sets, and with 80 or more training engines a ten-seed ensemble of the observation-corrected encoder is tied with the best simple or supervised HI. The practical value of the bottleneck encoder, which requires ten trainings of a recurrent model to reach a tie, is therefore not established, whereas the theory does not depend on this outcome.

Limitations. (1) The theory of Section 5 is local and Gaussian, and Theorem 3 assumes a constant-drift Wiener future; non-Gaussian posteriors, non-constant drift and regime switching are open, and the entropy estimates in the numerical checks have unquantified Monte Carlo error. (2) The efficiency η_{info} could not be estimated reliably on real data; the certificate is vacuous for most small data sets and, even on turbofan engines, is a lower bound that depends on the predictor and on a lifetime law that must be validated by a negative control, which failed for the test population of FD003. (3) The regime-aware variant and the variance study were designed after seeing results on the sets used, and no independent multi-regime data set was available (evidence tiers in Appendix A). (4) Ensembles of ten encoders are compared with deterministic simple HIs, which favours the encoders in computation. (5) The age fusion and the out-of-sample calibration were introduced after an initial protocol showed under-coverage; they benefit most HIs but are a pragmatic device, not a Bayes-exact fusion, and they mask part of the differences between HIs. The information that an HI adds to the age could not be certified with so few units (Section 7). (6) The bearing features are generic time-domain and spectral summaries, and bearing degradation is heterogeneous; physics-based features may change the picture. (7) The battery and bearing sets have three to seven units, and the benchmark of C-MAPSS is evaluated at the last observed cycle of each test engine, not along the test trajectories; the real-data tests of Section 8 use cross-validation within the data sets, cover simulated turbofan engines and two collections of lithium iron phosphate/graphite cells with charging-phase features, and include no bearing data set with more than a handful of units. (8) Proposition 5 concerns a scalar sensor and a regime-blind linear read-out; it does not extend to nonlinear multi-sensor estimators, and Theorem 4 concerns scalar linear HIs with fixed weights. The extension to snapshot-level projection limits [54] is left for future work.

Conclusions. Viewing HI construction as an information budget yields a score that can be computed before any predictor is trained, a certificate that can be computed after one is, and four structural results on what squeezing, scalar projection, regime blindness and integration lose. On batteries, bearings and turbofan engines, the predicted effects are observed, but no learned encoder is superior to simple indicators. The contribution of the paper is the budget, not a new predictor.

Acknowledgments
Acknowledgements are withheld for double-anonymous review.

Funding
Funding information is withheld for double-anonymous review.

Author contributions
Author contributions are withheld for double-anonymous review.

Data availability
NASA PCoE, NASA IMS, C-MAPSS, CALCE CS2 and FEMTO-ST/PRONOSTIA data are publicly available (NASA Prognostics Data Repository, CALCE Battery Group, IEEE 2012 Prognostics and Health Management data challenge). The simulation and benchmark code, the stored results and the per-unit outputs used to generate all tables and figures are available in an anonymised repository:
https://anonymous.4open.science/r/information-budget-health-indicator-rul-C5CE/.

A Data, protocols and evidence tiers

Data. Public run-to-failure collections were used: NASA PCoE cells B0005, B0006, B0007 and B0018 [72]; CALCE CS2 cells [73]; the FEMTO-ST/PRONOSTIA bearings [74]; the NASA IMS bearings [75, 76]; and the C-MAPSS turbofan engines [77]. For batteries, the capacity is used only to define the EOL and is never an input. The EOL is the first cycle at which the rolling-median

capacity falls to 75% (NASA) or 80% (CALCE) of the initial capacity; an absolute threshold of 1.4 Ah was not used because cell B0007 never reaches it. The lifetimes are 127, 73, 162 and 100 cycles (NASA) and 542, 499, 572 and 600 cycles (CS2_35 to CS2_38); the data are truncated at the EOL, and CALCE features are averaged over blocks of 4 cycles. Six per-cycle features are extracted from the charge and discharge records: constant-current time, constant-voltage time, height and voltage of the incremental-capacity peak, mean discharge temperature (NASA) or voltage drop at discharge onset (CALCE), and the discharge voltage at a fixed time (600 s for NASA, 1200 s for CALCE).

Further cells were screened by inclusion rules fixed before running any model (scripts and outputs are supplied). A NASA cell had to be at 24°C, have at least 100 discharge cycles, be discharged at a roughly constant 1.4 to 2.1 A, start between 1.5 and 2.1 Ah and reach the 75% EOL; only the four cells above qualify. Among the CALCE cells, CS2_5, 24 and 25 are partial-discharge tests, CS2_9 is discharged at a rate of 2C (twice the nominal capacity per hour), and CS2_3, 6 and 7 are short or follow other protocols, so none was used. CS2_33 and CS2_34 have the same chemistry and run to the EOL (485 and 479 cycles) but are discharged at 0.55 A; they are used only in a separate six-cell experiment, labelled protocol-heterogeneous, and the four-cell set remains the primary one.

For bearings, each run ends at failure, so EOL is the last snapshot. Condition 1 (nominally 1800 rpm, 4000 N; Bearing1_1–1_7) and condition 2 (1650 rpm, 4200 N; Bearing2_1–2_7) are evaluated separately, each with 7 units from the learning and the full test sets; condition 3 has only 3 units and was not used. The lifetimes are 2803, 871, 2375, 1428, 2463, 2448, 2259 snapshots (condition 1) and 911, 797, 1955, 751, 2311, 701, 230 snapshots (condition 2), each snapshot covering 0.1 s of 25.6 kHz acceleration every 10 s. Eight features are computed per snapshot: logarithm of the root mean square (RMS), kurtosis, crest factor and spectral entropy of the horizontal and of the vertical channel. Features are averaged over blocks of 20 (condition 1) or 8 (condition 2) snapshots, and the RUL is reported in snapshots.

Evaluation is leave-one-unit-out: each unit in turn is held out, the remaining units are used for training (three for NASA and four-cell CALCE, five for six-cell CALCE, six for each bearing condition), and metrics are computed at every step in the window 40–90% of life and pooled over the held-out units.

Further collections. FEMTO condition 3 (1500 rpm, 5000 N; Bearing3_1–3_3, with 515, 1637 and 434 snapshots, block length 10) and the NASA IMS bearings were kept aside and used only as confirmation sets. For IMS the four bearings that failed (Test 1 bearings 3 and 4, Test 2 bearing 1, Test 3 bearing 3; 2156, 2156, 984 and 6324 snapshots of 1 s at 20 kHz every 10 min) are used with one accelerometer channel each and four features per snapshot (log RMS, kurtosis, crest factor, spectral entropy), averaged over blocks of 20 snapshots. For C-MAPSS we use the single-condition subsets FD001 and FD003 (100 training engines run to failure and 100 truncated test engines with the official RUL labels each); the 14 informative sensors are standardised with training statistics. The 100 training engines are split by identifier into 80 engines to fit the HI constructions and 20 (identifier divisible by five) to select the bottleneck configuration and to calibrate the predictor; the test engines are predicted once at their last observed cycle, and 95% bootstrap intervals are computed over engines. The multi-regime subsets FD002 (260 training and 259 test engines) and FD004 (249 and 248) have six operating regimes; for them the only change is that sensors are standardised within regime, the six regimes being the clusters of the standardised operating settings found on the fitting engines (the clusters recover the six discrete settings exactly), the same preprocessing being applied to every method; the split is by identifier as above, with 208 and 200 fitting engines.

Methods. To isolate the effect of HI construction, every HI is mapped to the same scale and passed to the same calibrated predictor of Section 6.4. The sensor features are standardised, and each HI is mapped by an affine map $H \mapsto (H - H_\mathrm{s})/(H_\mathrm{e} - H_\mathrm{s})$, where H_s and H_e are the averages, over the *training* units only, of the mean of the first five and of the last five HI values of a unit (the end of life of a training unit is known); the standardisation, the PCA direction, the combination weights of the Geo-opt HI, the encoders and the affine map are all fitted on the training units and applied unchanged to the test unit, whose own end-of-life value is never used. The PGIB scale is set by its end-of-life penalty on the training units. The predictor of Section 6.4 then uses: the mixed inverse-Gaussian law (21) with the drift taken from the slope of the smoothed HI over a trailing window of $\max(10, 0.25t)$ steps, pooled with the age law. The parameters $(c, w_\mathrm{h}, w_\mathrm{a})$ are chosen on HI paths that are out of sample for the HI construction (leave-one-training-unit-out for

Table 5. Algorithm 1: training of the PGIB and Bayesian FHT prognostics (Section 6).

Input: training sequences $\{\mathbf{X}_{1:T_n}^{(n)}, T^{(n)}\}$, threshold $D_{\text{fail}} = 1$, compression weight β, step Δt. **Output:** q_{θ^*}, $(\vartheta^*, \sigma_B^*)$, predictive law (21).
1. Causal encoder (temporal convolutional network or GRU): $u_\theta, s_\theta \leftarrow \text{softplus}(\cdot)$ from $\mathbf{X}_{1:k}$.
2. Sample the path by reparametrisation: $H_k = H_{k-1} + u_\theta \Delta t + s_\theta \sqrt{\Delta t}\, \epsilon_k$, $\epsilon_k \sim \mathcal{N}(0,1)$, $H_0 = 0$.
3. Conjugate drift update from $\Delta H_{1:t}$ to obtain $(\hat{\mu}_t, \sigma_{\mu,t}^2)$.
4. $\mathcal{L}_{\text{FHT}} = -\ln p(T_t \mid H_{1:t})$ from (21); $\mathcal{L}_{\text{KL}}$ from Proposition 6.
5. Update $(\theta, \vartheta, \sigma_B)$ with the AdamW optimiser on $\mathcal{L}_{\text{FHT}} + \beta \mathcal{L}_{\text{KL}}$ (plus a penalty on $H_{T_{\text{EOL}}} - D_{\text{fail}}$ at the EOL).

the small sets, the 20 calibration engines for C-MAPSS). The compared HIs are PCA-HI (first principal component); the Geo-opt HI (linear combination found by differential evolution to maximise the sum of Mon, Tre and Pro on the training units); for C-MAPSS, the ridge HI (supervised linear regression of the life fraction on the features); a β-VAE (causal gated recurrent unit (GRU) encoder, one-dimensional latent variable, Gaussian prior); VIB (the PGIB with a zero-drift prior, that is, without physics); the PGIB (state-dependent drift $\eta(H;\vartheta) = \vartheta_1 \exp(\vartheta_2 H)$); and PGIB-KF (Section 6.3). For the bottleneck models, the compression weight $\beta \in \{0.03, 0.3, 3, 30\}$ and the encoder (GRU or causal feed-forward window network) are selected by inner leave-one-training-unit-out validation of the predictive negative log-likelihood, with at most three inner folds when more than five training units are available; for PGIB-KF only $\beta \in \{0.03, 0.3, 3\}$ is searched. The held-out unit is never used. Learned models use three seeds, and the other HIs are deterministic. The sensor-free reference is the age law p_{age} of Section 6.4. The metrics are the RMSE and the mean absolute error (MAE) of the median prediction, the prediction interval coverage probability (PICP) and normalised average width (PINAW) of the 95% interval, and the continuous ranked probability score (CRPS) [66]. For C-MAPSS the asymmetric score is $\sum_i (e^{-d_i/13} - 1)$ for $d_i < 0$ and $\sum_i (e^{d_i/10} - 1)$ for $d_i \geq 0$, summed over engines, with d_i the predicted minus the true RUL [77].

Protocol history and evidence tiers. The study proceeded in five protocols, which determine how the numbers can be read. P1 (Appendix C) was the first benchmark; it showed that the integrating encoder under-covers and that the age law is competitive. P2 introduced the resulting improvements (out-of-sample calibration, age fusion, a prediction grid independent of the test unit) and was frozen, with the file hash recorded, before it was run on the two confirmation sets (FEMTO condition 3 and IMS) and before the development sets were re-run. On C-MAPSS FD001 the P2 methods showed that the integrating encoder is considerably worse than simple HIs even with 80 training engines. The diagnosis of Proposition 7 led to P3, the observation-corrected PGIB-KF, which was frozen and run once on FD003, a subset not used before, and exploratorily on the small sets. The unchanged P3 model was then frozen with regime-wise preprocessing and run once on FD002 and FD004. (A first launch used only the first 100 training engines because the identifier list was hard-coded in the script; it was stopped before any result was produced, and the script and the recorded hash were corrected.) After these results, P4 added the regime-aware encoder and a regime-specific ridge baseline and was run once on each subset, the choice between PGIB-KF and PGIB-KFR and of β being made on the calibration engines only. Because FD002 and FD004 had already been seen, they are development sets for P4, and no independent multi-regime data set was available (the successor N-CMAPSS has only six to ten units per subset). Finally, because the differences between learned encoders and simple HIs were of the size of the seed-to-seed variability, P5 retrained the learned encoders on the four subsets with ten fresh seeds, keeping architecture, β and calibration fixed; it supersedes the three-seed results where they disagree. The evidence tiers are therefore: (i) *clean confirmation*: C-MAPSS FD003, FD002 and FD004 for P3 (not for P4), and FEMTO condition 3 and IMS for P2 (not for P3, whose idea followed their P2 results); (ii) *development*: NASA, CALCE (four and six cells), FEMTO conditions 1 and 2 and C-MAPSS FD001, which informed the design and whose results are therefore not unbiased estimates of out-of-sample performance.

B Detailed benchmark results

This appendix gives the tables behind Section 7 with a short commentary; the numerical details are in the tables.

Few training units, development sets. No method is best everywhere (Table 7). The Geo-opt HI is best on NASA and six-cell CALCE, the PGIB on four-cell CALCE (38.4 against 43.3 for the

best simple HI and 46.5 for the age law), and the age law on both FEMTO conditions. On NASA no learned HI beats the age law. The PGIB-KF, designed for many training units, is clearly worse than the PGIB with so few units.

Confirmation sets for P2. On the two sets not used in the design (Table 8), the β-VAE is best on FEMTO condition 3 and PCA-HI on IMS, the latter by a wide margin, and the PGIB is only 4% better than the age law on IMS. The bottleneck encoder is therefore not superior in the few-unit regime on unseen data.

Many training units: C-MAPSS. On FD003, the clean confirmation set of P3 (Table 9), a three-seed ensemble of PGIB-KF had the lowest RMSE, 19.4 (95% interval 16.0 to 22.8), against 21.9 for the Geo-opt HI, 26.8 for the ridge HI, 33.8 for PCA-HI, 39.5 for the integrating PGIB and 47.6 for the age law; the paired difference from the Geo-opt HI is -2.5 (-4.8 to -0.3). A single model has RMSE 24.2, and the ten-seed study shows that the three-seed result was favourable. On FD001 (Table 10) the PGIB-KF (22.9) is indistinguishable from the best simple HIs (21.8 to 22.2), with the lowest CRPS and narrower intervals. The integrating encoder is considerably worse on both subsets, and removing the error accumulation brings the same objective from the level of the age law to that of the best simple HIs or better.

Multi-regime confirmation: FD002 and FD004. The FD003 result does not replicate on the multi-regime subsets, which were frozen and run once (Tables 11 and 12). On FD002, PCA-HI is best (28.7) and PGIB-KF is slightly but significantly worse (+1.6, interval 0.3 to 3.0). On FD004, the Geo-opt HI (29.7) and the ridge HI (30.9) are well ahead of PGIB-KF (40.6), which still beats PCA-HI, the β-VAE, the integrating PGIB and the age law but under-covers (coverage 81% on FD002 and 79% on FD004).

Regime-aware observation model (P4). Adding the regime to the observation model (PGIB-KFR) narrows the gap on the multi-regime subsets but does not close it. On FD002 the PGIB-KFR (29.6) is indistinguishable from PCA-HI, the Geo-opt HI and the regime-specific ridge HI (28.4, the lowest). On FD004 it improves from 40.6 to 34.8 and beats PCA-HI (-9.5, -12.4 to -6.6) but stays behind the Geo-opt HI (+5.1, 2.3 to 7.9) and the ridge HI. Two points limit the reading. The pre-declared selection by calibration likelihood chose PGIB-KF rather than PGIB-KFR on three of the four subsets, and the single-regime check failed: with one regime, PGIB-KFR is PGIB-KF up to an extra input and the initialisation, yet it is worse on FD001 (26.5 against 22.9) and FD003 (25.4 against 19.4). Differences of three to six RMSE units between near-equivalent configurations are within the training variability. We therefore do not claim that the regime-aware model improves prognostics; the result is consistent with Proposition 5 but does not demonstrate it.

Seed variability and ten-seed ensembles (P5). The learned models were retrained with ten fresh seeds each, keeping architecture, β and calibration fixed (Table 4, Figure 7). The ensembles have a coverage of 84% to 94%. PGIB-KFR is worse than PGIB-KF on FD001 by 3.7 RMSE units (Welch test [78], $p = 0.02$) and gains only with six regimes on FD004 (-4.0, $p = 0.17$).

Ablations and fusion. On FD003, removing the bottleneck ($\beta = 0$) or the physics prior changes the RMSE of the integrating PGIB by less than two units (39.1 and 37.9 against 39.9), whereas removing the epistemic drift variance raises it to 45.9 and lowers the coverage from 95% to 92%. The age fusion (Table 13) improves the point error of most HIs when the HI is poor or the units are few; for example, the RMSE of PCA-HI on FD001 falls from 29.0 to 21.8.

Information certificates. The certified information (Table 14) is vacuous for NASA, the three FEMTO conditions and, except for PCA-HI, IMS. It is positive only on CALCE, where it is 0.60 nats for the PGIB against 0.41 for the age law with four cells, and 0.35 against 0.31 with six cells. These are lower bounds on $I(T_t; H_{1:t})$ and on $I(T_t; t)$ respectively, so their difference does not bound the information added by the HI. To separate the two, Proposition 2 gives $I(T_t; H_{1:t} \mid t) \geq \bar{h}(T_t \mid t) - \frac{1}{2}\ln(2\pi e\,\mathrm{MSE})$, where $\bar{h}(T_t \mid t)$ is the mean entropy of the RUL given the age. We evaluated it under a log-normal lifetime model fitted to all lifetimes of each data set (Table 6). The age-only predictor cannot exceed the entropy-power bound of the true age-conditional law, so its value must not be positive; it is positive on CALCE (four and six cells)

Table 6. Age-conditional certificate (nats) under a fitted log-normal lifetime model; $\bar{h}(T \mid t)$: mean conditional entropy. A positive value for the age-only predictor shows that the model overestimates $h(T \mid t)$.

Data set	$\bar{h}(T \mid t)$	Age-only	PCA	Geo-opt.	β-VAE	VIB	PGIB
NASA	4.67	-0.28	0.04	0.07	-0.14	-0.23	-0.29
CALCE-4	5.74	0.49	0.39	0.56	0.42	0.46	0.68
CALCE-6	5.69	0.35	0.23	0.44	-0.05	0.31	0.38
FEMTO-1	7.81	0.19	-0.28	0.11	-0.15	-0.03	0.11
FEMTO-2	7.77	0.29	0.03	0.13	0.08	0.25	0.24
FEMTO-3	7.30	-0.45	-0.35	-0.46	-0.19	-0.34	-0.35
IMS	8.65	-0.20	0.62	0.39	-0.58	-0.31	-0.16

and on FEMTO conditions 1 and 2, because a log-normal law fitted to three to seven lifetimes overestimates $h(T_t \mid t)$. The model-based age-conditional certificate therefore fails this negative control and is not used to claim that any HI adds information to the age.

C Initial benchmark protocol (P1)

The first benchmark used a simpler protocol (P1). Every HI was calibrated on in-sample training paths, the plain first-hitting-time law (21) was used without age fusion, and the predictive grid extended to three times the held-out unit's own life, which leaks the length of the test unit into the support of the predictive law (removed in P2). The pilot on NASA B0005, the results below and the observation that the age law is competitive motivated P2. The P1 results are reported for completeness and are not used for the conclusions of the main text.

Pilot run. A pilot on the NASA B0005 fold, with a GRU-only encoder and $\beta \in \{0.003, 0.03, 0.3\}$, gave a PGIB RMSE of 56.3 cycles and zero interval coverage, with the selected β at the edge of the grid. The selection grid was therefore extended and the feed-forward encoder added as a candidate. Both changes are chosen per fold by inner validation, but the decision to introduce them was made after seeing a held-out result for one fold. No baseline was modified, and no change was made after the full runs reported below; the bearing experiments were run once with the final protocol.

Results under P1. The outcome depends on the data set (Tables 15 and 16), and the age law is needed to read it. On NASA the age-law RMSE is 39.8 cycles; PCA-HI (17.8), the Geo-opt HI (21.9), the β-VAE (30.8) and VIB (33.3) beat it, whereas the PGIB (47.6) does not, and its intervals cover only 42% of the true RULs. On CALCE the age-law RMSE is 49.3; only the PGIB (33.0) beats it clearly, while PCA-HI, the Geo-opt HI and the β-VAE (117 to 164) are considerably worse than the age law. With the two 0.5C-discharge cells added, the PGIB keeps the lowest RMSE (50.4) but its margin over the age law (54.7) is only 8%. On bearings the age-law RMSE (602 and 664 snapshots for conditions 1 and 2) is comparable to the standard deviation of the pooled RUL, and most HIs do no better: the PGIB has the lowest RMSE in condition 1 (524.5, 13% below the age law) but gives almost no benefit in condition 2 (621.8 against 664). Two long-lived units dominate the pooled error in condition 2.

The ablations (in the supplied result files) support the bottleneck on all five sets (for example RMSE $54.4 \to 47.6$ on NASA and $42.3 \to 33.0$ on CALCE) and the epistemic drift variance for coverage (for example $14 \to 42\%$ on NASA). The physics prior helps on CALCE and FEMTO condition 1 but not on NASA or FEMTO condition 2. With only three to six training units per fold, the validation signal that selects the encoder is itself noisy; on NASA it chose the strongest compression ($\beta = 30$), at the edge of the grid, in all eight selections.

Geometric scores versus error. Across the five HIs of each data set, the rank correlation between the geometric score Mon + Tre + Pro and the RMSE is +0.90 on NASA (a higher geometric score goes with a larger error), −1.00 on CALCE, −0.90 on six-cell CALCE, −0.60 on FEMTO condition 1 and +0.60 on condition 2. The geometric criteria therefore did not rank the HIs consistently by prognostic error. This is an association over five methods per data set and not evidence for the mechanism of Theorem 2: the VIB and PGIB HIs have Mon $\approx$ Tre ≈ 1 because their velocity is constrained to be positive, not because of a squeeze.

Information on real data. A direct estimate of $I(T; H)$ with the Kraskov–Stögbauer–Grassberger estimator [48] on rank-transformed window summaries gave, for several HIs, values above the estimate for the 12-dimensional raw sensor summary, which is impossible for a true mutual information because the summary contains the HI. This reflects the

Table 7. Protocol P2, development sets. RMSE, MAE and CRPS in cycles (batteries) or 10 s snapshots (bearings); mean ± standard deviation over seeds; bold: lowest RMSE.

NASA PCoE (4 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	34.4	28.7	84	2.68	21.0
PCA-HI	24.9	19.3	95	2.46	13.8
Geo-opt HI	**24.3**	20.1	93	3.22	17.6
β-VAE-HI	31.7±2.2	25.3±1.8	98±0	4.24±0.17	20.0±0.9
VIB-HI (no physics)	33.9±1.2	28.3±1.2	90±1	5.40±0.03	27.1±0.7
PGIB-HI (ens.)	34.8	29.0	93	6.51	27.3
PGIB-KF-HI (ens.)	44.3	39.4	84	4.55	29.1

CALCE CS2 (4 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	46.5	40.0	100	1.13	28.6
PCA-HI	51.3	44.0	100	1.06	30.2
Geo-opt HI	43.3	35.7	98	0.98	26.5
β-VAE-HI	49.9±0.9	41.6±1.0	100	1.09±0.00	29.5±0.4
VIB-HI (no physics)	48.3±1.4	40.6±0.9	83±4	0.85±0.00	30.8±0.8
PGIB-HI (ens.)	**38.4**	34.2	99	0.74	23.3
PGIB-KF-HI (ens.)	50.3	36.6	77	0.83	33.0

CALCE CS2 (6 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	50.6	45.5	100	1.13	30.1
PCA-HI	56.8	50.0	100	1.09	32.8
Geo-opt HI	**46.3**	43.2	100	1.02	27.6
β-VAE-HI	73.5±4.9	60.9±3.9	93±3	1.04±0.02	42.4±3.2
VIB-HI (no physics)	54.7±1.7	48.7±1.5	92±1	0.91±0.01	32.4±1.2
PGIB-HI (ens.)	48.9	44.8	90	0.87	30.8
PGIB-KF-HI (ens.)	71.9	63.6	84	0.92	45.2

FEMTO cond. 1 (7 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	**495.8**	371.4	94	3.45	303.7
PCA-HI	786.9	621.6	72	4.72	489.2
Geo-opt HI	533.2	416.8	96	3.19	311.5
β-VAE-HI	737.1±191.7	525.4±102.0	85±6	4.84±0.04	421.4±87.9
VIB-HI (no physics)	584.1±29.4	467.0±26.4	82±5	3.48±0.13	357.9±29.0
PGIB-HI (ens.)	534.7	401.8	84	4.67	340.9

FEMTO cond. 2 (7 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	**430.6**	373.5	97	10.38	278.6
PCA-HI	558.3	450.5	67	8.52	391.1
Geo-opt HI	504.9	414.7	85	11.22	396.1
β-VAE-HI	531.2±11.3	434.6±20.1	88±2	9.78±0.02	368.4±2.0
VIB-HI (no physics)	448.2±0.4	337.7±1.3	97	11.39±0.14	318.5±2.6
PGIB-HI (ens.)	449.5	333.6	97	13.97	341.2

Table 8. Protocol P2, confirmation sets (snapshots). PGIB-KF is exploratory here.

FEMTO cond. 3 (3 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	563.6	529.7	37	10.60	460.2
PCA-HI	507.5	418.4	32	6.50	398.8
Geo-opt HI	567.1	525.1	30	12.82	453.4
β-VAE-HI	**462.0±20.8**	389.5±23.6	37	8.86±0.72	363.4±20.9
VIB-HI (no physics)	505.0±0.1	438.3±0.3	37	12.55±0.00	442.5±0.2
PGIB-HI (ens.)	510.3	450.0	37	12.56	453.5
PGIB-KF-HI (ens.)	555.0	479.3	33	9.10	438.6

NASA IMS (4 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	1692.8	1503.1	83	10.78	1106.3
PCA-HI	**745.2**	605.7	96	8.05	498.0
Geo-opt HI	942.7	702.8	99	16.31	771.6
β-VAE-HI	2769.6±282.2	1324.8±242.7	84±6	10.31±0.71	1184.6±197.8
VIB-HI (no physics)	1884.6±10.5	1434.4±12.4	91	16.41±0.01	1122.5±8.8
PGIB-HI (ens.)	1623.8	1288.1	98	16.49	1017.5
PGIB-KF-HI (ens.)	1852.3	1519.8	43	12.29	1545.6

Table 9. C-MAPSS FD003 (clean confirmation set of P3; PGIB-KFR and regime-specific ridge rows are development results). RMSE: single models, mean over seeds; seed-avg.: RMSE of the seed-averaged prediction with 95% bootstrap interval over engines.

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	47.6	42.3–52.6	39.6	114536	99	1.91	28.6
PCA-HI	33.8	28.4–38.3	25.5	10747	97	1.47	19.7
Geo-opt HI	21.9	17.8–26.0	15.9	1864	100	1.19	13.1
Ridge-HI (supervised)	26.8	22.9–30.6	21.0	6147	93	1.20	16.2
Ridge-HI, regime-specific	26.8	22.8–30.6	21.0	6147	93	1.20	16.2
β-VAE-HI	40.6±1.1	34.9–45.6	32.0±0.7	29236	99	1.69	23.8±0.3
VIB-HI (no physics prior)	37.9±0.8	31.7–42.2	29.3±0.8	18680	96	2.16	24.2±0.6
PGIB-HI (GRU/window encoder)	39.9±1.0	33.9–44.4	31.2±0.4	21945	95	1.98	24.3±0.3
PGIB-HI, 3-seed ensemble	39.5	34.1–45.0	31.1	17643	95	1.98	24.1
PGIB-KF-HI	24.2±2.5	16.4–23.1	18.6±1.5	2296	94	1.19	14.8±0.4
PGIB-KF-HI, 3-seed ensemble	19.4	16.0–22.8	14.6	790	97	1.20	12.2
PGIB-KFR-HI (regime-aware)	27.4±0.1	21.8–29.3	21.2±0.3	5417	95	1.24	15.6±0.4
PGIB-KFR-HI, 3-seed ensemble	25.4	21.6–29.1	20.3	4170	97	1.24	14.6

Table 10. C-MAPSS FD001 (development set; same layout as Table 9).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	36.6	32.4–40.8	30.5	8023	99	1.18	21.1
PCA-HI	21.8	19.1–24.4	17.9	908	96	0.83	12.6
Geo-opt HI	22.0	19.5–24.3	18.0	953	96	0.83	12.7
Ridge-HI (supervised)	26.0	22.0–29.8	19.7	1606	97	0.85	14.1
Ridge-HI, regime-specific	26.0	22.0–29.8	19.7	1606	97	0.85	14.1
β-VAE-HI	22.2±0.4	19.0–24.5	18.0±0.5	875	95	0.82	12.7±0.3
VIB-HI (no physics prior)	34.2±0.3	28.8–38.3	27.1±0.7	7386	91	0.96	19.3±0.4
PGIB-HI (GRU/window encoder)	35.6±1.0	29.8–40.9	26.9±0.6	11524	94	0.90	19.6±0.4
PGIB-HI, 3-seed ensemble	35.4	29.9–40.9	26.6	10874	94	0.90	19.4
PGIB-KF-HI	27.0±2.2	18.8–27.1	20.2±1.2	2868	82	0.56	14.9±1.1
PGIB-KF-HI, 3-seed ensemble	22.9	18.6–26.9	16.6	1309	93	0.56	12.1
PGIB-KFR-HI (regime-aware)	26.9±3.6	21.8–31.0	20.2±2.6	2937	94	0.64	14.4±2.1
PGIB-KFR-HI, 3-seed ensemble	26.5	22.0–30.9	19.5	2094	96	0.64	13.8

Table 11. C-MAPSS FD002 (multi-regime; confirmation set for P3, development set for P4; same layout as Table 9).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	38.3	35.2–41.2	31.4	53131	97	0.82	21.6
PCA-HI	28.7	26.0–31.4	21.9	10331	95	0.66	15.7
Geo-opt HI	30.5	27.0–33.9	21.8	16164	98	0.59	15.2
Ridge-HI (supervised)	29.4	26.1–32.6	21.7	11935	94	0.58	15.1
Ridge-HI, regime-specific	28.4	25.3–31.5	20.9	11149	94	0.58	14.7
β-VAE-HI	30.6±0.8	27.3–33.1	24.0±0.7	11996	92	0.65	17.0±0.4
VIB-HI (no physics prior)	36.4±0.4	33.3–39.4	29.8±0.2	26118	82	0.68	21.1±0.1
PGIB-HI (GRU/window encoder)	36.8±0.2	33.6–39.4	30.1±0.5	28553	83	0.68	21.4±0.5
PGIB-HI, 3-seed ensemble	36.6	33.7–39.4	29.9	26625	82	0.68	21.3
PGIB-KF-HI	31.3±1.5	27.8–33.1	25.4±1.5	14139	81	0.79	18.6±1.1
PGIB-KF-HI, 3-seed ensemble	30.4	27.8–32.9	24.5	10983	81	0.79	18.0
PGIB-KFR-HI (regime-aware)	30.7±2.0	27.2–32.2	24.8±2.6	14802	86	0.91	18.5±1.9
PGIB-KFR-HI, 3-seed ensemble	29.6	27.0–32.1	23.7	13089	87	0.91	17.8

Table 12. C-MAPSS FD004 (multi-regime; confirmation set for P3, development set for P4; same layout as Table 9).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	54.7	50.9–58.5	46.0	611896	95	1.31	31.8
PCA-HI	44.3	40.7–48.0	35.4	104864	92	1.03	24.9
Geo-opt HI	29.7	26.6–32.7	21.9	9468	100	0.86	16.1
Ridge-HI (supervised)	30.9	28.1–33.4	23.8	11632	94	1.14	18.5
Ridge-HI, regime-specific	31.1	28.2–33.9	24.0	10794	92	1.14	18.6
β-VAE-HI	48.1±3.1	42.7–50.2	39.0±3.0	294449	93	1.17	27.7±1.9
VIB-HI (no physics prior)	51.8±1.0	47.3–56.2	41.9±0.6	697542	90	1.13	29.5±0.4
PGIB-HI (GRU/window encoder)	49.9±2.4	44.2–53.9	39.5±1.5	1143425	90	1.42	29.3±0.7
PGIB-HI, 3-seed ensemble	49.5	44.8–54.3	38.9	1085286	91	1.42	28.9
PGIB-KF-HI	42.1±2.8	37.2–43.6	33.9±1.9	241870	78	1.08	25.6±1.3
PGIB-KF-HI, 3-seed ensemble	40.6	37.3–44.0	32.9	124580	79	1.07	24.7
PGIB-KFR-HI (regime-aware)	37.1±2.5	31.4–37.2	29.4±1.5	187056	85	1.12	22.3±0.8
PGIB-KFR-HI, 3-seed ensemble	34.8	31.8–37.8	27.8	37063	85	1.12	21.0

Table 13. Effect of the age fusion: RMSE with fusion / without fusion (after recalibration with $w_a = 0$).

HI	NASA	CALCE-4	CALCE-6	FEMTO-1	FEMTO-2	FEMTO-3	IMS
PCA-HI	25 / 18	51 / 186	57 / 184	787 / 1470	558 / 564	507 / 503	745 / 745
Geo-opt HI	24 / 32	43 / 131	46 / 126	533 / 1083	505 / 489	567 / 437	943 / 943
β-VAE-HI	32 / 26	50 / 110	73 / 270	737 / 1421	531 / 457	462 / 396	2770 / 2770
VIB-HI (no physics)	34 / 30	48 / 59	55 / 106	584 / 681	448 / 469	505 / 507	1885 / 1885
PGIB-HI (ens.)	35 / 30	38 / 35	49 / 85	535 / 754	449 / 445	510 / 509	1624 / 1624
PGIB-KF-HI (ens.)	44 / 44	50 / 59	72 / 88	–	–	555 / 556	1852 / 1852

Table 14. Protocol P2, information certified by the achieved error (nats); < 0: vacuous. $h(T)$: entropy of the pooled held-out RUL.

Dataset	$h(T)$	Age-only	PCA	Geo-opt.	β-VAE	VIB	PGIB	PGIB-KF
NASA	4.36	< 0	< 0	< 0	< 0	< 0	< 0	< 0
CALCE-4	5.67	0.41	0.31	0.48	0.34	0.37	0.60	0.33
CALCE-6	5.66	0.31	0.20	0.40	< 0	0.24	0.35	< 0
FEMTO-1	7.24	< 0	< 0	< 0	< 0	< 0	< 0	–
FEMTO-2	7.04	< 0	< 0	< 0	< 0	< 0	< 0	–
FEMTO-3	6.70	< 0	< 0	< 0	< 0	< 0	< 0	< 0
IMS	8.05	< 0	0.02	< 0	< 0	< 0	< 0	< 0

Table 15. Protocol P1, battery benchmarks, leave-one-unit-out (mean ± standard deviation over seeds). RMSE, MAE and CRPS in cycles.

NASA PCoE (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	39.8	34.7	–	–	–
PCA-HI	0.27	0.85	0.77	17.8	14.0	97	2.52	11.1
Geo-opt HI	0.21	0.93	0.95	21.9	17.0	90	2.43	12.8
β-VAE-HI	0.30±0.01	0.71±0.00	0.95±0.01	30.8±0.8	26.0±0.5	95±1	2.06±0.08	17.6±0.5
VIB-HI (no physics prior)	1.00	1.00±0.00	0.67±0.03	33.3±2.8	27.7±2.1	88±4	3.47±0.10	20.5±2.3
PGIB-HI	1.00	1.00±0.00	0.67±0.01	47.6±2.1	42.3±1.9	42±3	2.39±0.15	34.3±1.5
CALCE CS2 (4 cells, 1C/1C) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	49.3	43.8	–	–	–
PCA-HI	0.05	0.72	0.83	130.5	110.4	92	2.74	82.3
Geo-opt HI	0.39	0.95	0.98	117.2	86.8	90	3.20	69.2
β-VAE-HI	0.02±0.01	0.72±0.01	0.80±0.01	163.7±0.7	148.0±0.9	83±1	1.87±0.02	111.1±0.7
VIB-HI (no physics prior)	1.00	1.00±0.00	0.92±0.01	53.2±0.9	47.2±1.4	71±4	1.24±0.13	38.4±0.7
PGIB-HI	1.00	1.00±0.00	0.95±0.01	33.0±4.9	29.9±3.9	55±8	0.34±0.02	22.6±3.1
CALCE CS2 (6 cells, incl. 0.5C-discharge CS2_33/34) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	54.7	50.0	–	–	–
PCA-HI	0.13	0.28	0.21	191.8	172.0	50	0.99	149.6
Geo-opt HI	0.39	0.95	0.98	106.9	82.7	95	3.11	63.7
β-VAE-HI	0.05±0.02	0.52±0.10	0.33±0.04	201.6±5.3	179.9±5.0	48±0	1.46±0.18	156.1±3.1
VIB-HI (no physics prior)	1.00	1.00±0.00	0.87±0.02	69.7±6.0	54.4±8.8	90±5	2.24±0.23	44.9±4.8
PGIB-HI	1.00	1.00±0.00	0.91±0.01	50.4±4.4	44.7±5.1	69±5	0.58±0.02	33.4±4.2

Table 16. Protocol P1, FEMTO bearing benchmarks. RMSE, MAE and CRPS in 10 s snapshots.

FEMTO bearings, condition 1 (7 units) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	602.5	515.4	–	–	–
PCA-HI	0.06	0.56	0.70	784.5	663.2	93	2.80	485.7
Geo-opt HI	0.04	0.29	0.39	770.2	623.3	91	3.34	459.1
β-VAE-HI	0.07±0.02	0.58±0.01	0.73±0.01	779.8±51.1	664.4±35.6	91±3	2.69±0.13	492.4±20.6
VIB-HI (no physics prior)	1.00	1.00±0.00	0.71±0.02	570.1±56.0	471.0±50.7	78±5	3.04±0.21	365.6±44.1
PGIB-HI	1.00	1.00±0.00	0.71±0.02	524.5±45.3	423.6±37.3	75±5	2.65±0.10	336.5±31.8
FEMTO bearings, condition 2 (7 units) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	664.4	584.6	–	–	–
PCA-HI	0.04	0.43	0.17	613.9	487.6	54	2.25	433.9
Geo-opt HI	0.06	0.37	0.46	559.2	457.1	73	2.33	356.5
β-VAE-HI	0.06±0.01	0.46±0.05	0.42±0.16	539.8±31.7	439.6±24.6	70±2	2.01±0.06	348.6±26.1
VIB-HI (no physics prior)	1.00	0.99±0.00	0.39±0.02	616.8±1.5	493.8±1.5	41±1	2.62±0.23	451.1±3.2
PGIB-HI	1.00	0.99±0.00	0.46±0.02	621.8±4.3	498.2±8.1	36±2	2.25±0.01	464.3±7.9

Table 17. Protocol P1, information certified by the achieved error (Proposition 1), in nats; < 0: vacuous. $h(T)$: entropy of the pooled held-out RUL.

Dataset	$h(T)$	PCA	Geo-opt.	β-VAE	VIB	PGIB
NASA	4.36	0.06	< 0	< 0	< 0	< 0
CALCE (4)	5.67	< 0	< 0	< 0	0.27	0.76
CALCE (6)	5.66	< 0	< 0	< 0	< 0	0.32
FEMTO c. 1	7.24	< 0	< 0	< 0	< 0	< 0
FEMTO c. 2	7.04	< 0	< 0	< 0	< 0	< 0

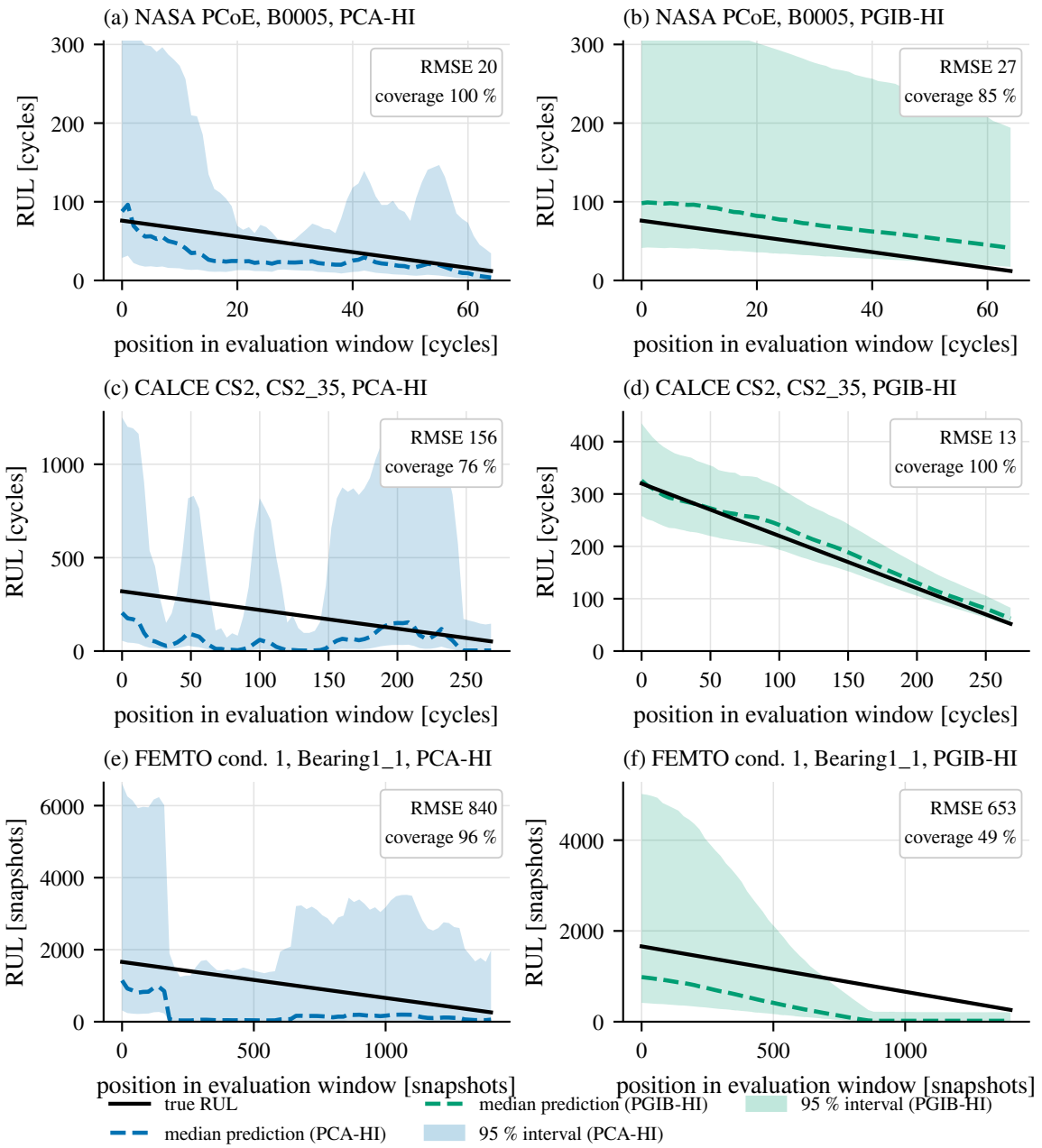


Figure 10. Example RUL predictions with 95% intervals on one held-out unit per data set (NASA B0005, CALCE CS2_35, FEMTO Bearing1_1), for PCA-HI (left) and PGIB (right); each panel gives the RMSE and the coverage on that unit.

bias of the estimator in 12 dimensions at a few hundred samples, a known limitation [49, 50]. A simulation with known mutual information (twelve Gaussian sensor features, a scalar target and a scalar HI with $\eta_{\text{info}} = 0.7$, Table 18) shows that the estimator underestimates $I(T;\mathbf{X})$ strongly, so that $\hat{\eta}_{\text{info}}$ exceeds 1 in every replication even with 3000 samples. We therefore do not report η_{info} on real data. The certificate of Proposition 1 avoids high-dimensional estimation (Table 17); it is positive only for the PGIB on CALCE (0.76 nats) and six-cell CALCE (0.32), VIB on CALCE (0.27) and PCA-HI on NASA (0.06), and vacuous otherwise. This is a more modest reading of the benchmark than the RMSE tables alone.

D Details of the real-data tests

Turbofan engines: set-up. Each subset uses its training engines (run to failure); the 14 informative sensors are standardised with the training engines of the fold, within regime for FD002 and FD004 (six regimes from k-means clustering of the operating settings). The ridge HI is a ridge regression (penalty 10) of the life fraction on the standardised sensors, smoothed by a trailing mean of 5 cycles and mapped affinely to zero at the start and one at the end of life with the training

Table 18. Estimation of η_{info} with the Kraskov–Stögbauer–Grassberger estimator ($k = 5$, 100 replications) in a Gaussian model with known mutual information; n is the sample size.

n	$\hat{I}(T;\mathbf{X})$	$\hat{I}(T;H)$	$\hat{\eta}_{\text{info}}$ (mean $\pm$ standard deviation)	fraction $\hat{\eta}_{\text{info}} > 1$
100	0.23	0.59	2.72 ± 0.74	1.00
300	0.33	0.68	2.10 ± 0.25	1.00
1000	0.43	0.71	1.65 ± 0.09	1.00
3000	0.51	0.71	1.40 ± 0.04	1.00

true values: $I(T;\mathbf{X}) = 1.00$, $I(T;H) = 0.70$, $\eta_{\text{info}} = 0.70$ (nats)

engines of the fold; PCA-HI is the first principal component. The trajectory summary of an HI consists of its trailing mean over 10 cycles (level), its least-squares slope over the last 30 cycles (velocity), its values 20 and 50 cycles earlier, and the age; the raw-sensor predictor uses level and slope of every sensor and the age. Predictors are histogram-based gradient-boosted regression trees (200 iterations, learning rate 0.08, 31 leaves, at least 40 samples per leaf, L_2 penalty 1) trained on all pairs of the training engines of a fold. The lifetime law is a Gaussian kernel density of $\ln L$ (Scott bandwidth) of the training engines of the fold; the conditional density of the RUL at age t is $f_L(t+\tau)/S_L(t)$ and $h(T_t \mid t)$ is evaluated numerically on a grid of 0.5 cycles. Bootstrap intervals resample engines (300 replicates).

Lithium-ion cells. Each cycle of the processed records yields the discharge capacity, the charge duration (samples with current above 0.01 A) and the charge voltage when the charged capacity first reaches 0.3 and 0.8 Ah; the HIs use the last three quantities. The charge duration is divided by its mean over cycles 2 to 11 of the same cell and the voltages are reduced by their means over these cycles, and the result is standardised with the training cells of the fold. Time scales are set by the lifetimes. For HUST the minimum age is 100 cycles, ages are sampled every 5 cycles, the windows are 20 cycles (level) and 100 cycles (velocity), the lags are 100 and 250 cycles and the grid of the conditional entropy has a step of 2 cycles. For MATR the corresponding values are 50, 3, 10, 50, 50 and 125 cycles and a step of 1 cycle. The trees have at most 15 leaves, and all other settings are those of the engines. The capacity reference uses level and velocity of the relative discharge capacity and the age. The end of life is the first cycle at which the running median of the discharge capacity over 5 cycles falls below 80% of the mean over the first 5 cycles (HUST), or the life label of the source, at which the recordings end (MATR; recordings that end more than 10 cycles before the label would be excluded, and none was). The processed records are unpickled with a restricted reader that admits only array types. The protocol read-out of Table 22 uses the three charging features standardised over the population (blind) or relative to the own cycles 2 to 11 (normalised), with the ridge read-out of the engines.

Squeeze. The template is the mean ridge HI at each cycle over the training engines of the fold, and the noise standard deviation is $\sigma_2 = \Delta m/(\sqrt{2} \cdot 1.645)$ with Δm the mean template increment over the first half of the median lifetime. Mon_∞ is predicted as the mean of $2\Phi_{\mathcal{N}}(\Delta m_k/(\sqrt{2}\sigma_2)) - 1$ over the template increments Δm_k up to the median lifetime, where $\Phi_{\mathcal{N}}$ is the standard normal distribution function. Because the template slope varies along life, the predicted limit is lower than for a constant slope.

Stages. Level and velocity are the summaries of the out-of-fold ridge HI, standardised over all pairs; the target is the RUL. Stages are bins of the life fraction (0.2 to 1, edges $0.2, 0.35, 0.5, 0.65, 0.8$; oracle stages) or equal-count bins of the age (observable stages). Within a stage, the free direction is the linear least-squares fit on both features, level-only and velocity-only use one feature, and a static direction (grid of 5°) is a fixed combination of the two features with a stage-specific affine map; the static direction minimises the sum of the out-of-fold MSEs over stages. The tables give the worst-stage RMSE ratio to the free direction.

Official test engines. The certificate uses the observed cycles of each test engine as its age, the lifetime law of the training engines, and the predictions of the methods of Section 7 (ten-seed ensembles for the learned encoders); the age law is the control. It assumes that the test truncation is independent of the lifetime, which the control checks.

Table 19. Cross-validated age-conditional certificates (nats, 95% bootstrap interval); $\bar{h}(T \mid t)$: mean conditional entropy. Sensors: raw sensors (engines) or charging features (cells); Capacity: reference predictor on the discharge capacity (cells).

Subset	Units	$\bar{h}(T \mid t)$	Age only (control)	Ridge HI	PCA-HI	Level only	Slope only	Sensors	Capacity (ref.)
FD001	100	5.11	-0.11 [-0.28, 0.08]	0.31 [0.20, 0.44]	0.25 [0.13, 0.37]	0.07 [-0.10, 0.24]	0.17 [0.01, 0.32]	0.22 [0.06, 0.38]	–
FD002	260	5.11	-0.11 [-0.22, -0.01]	0.21 [0.13, 0.30]	0.19 [0.11, 0.27]	0.11 [0.03, 0.21]	0.11 [0.01, 0.20]	0.16 [0.08, 0.25]	–
FD003	100	5.64	-0.22 [-0.37, -0.03]	0.10 [-0.05, 0.28]	0.08 [-0.07, 0.25]	0.06 [-0.09, 0.24]	0.00 [-0.13, 0.16]	0.20 [0.04, 0.40]	–
FD004	249	5.52	-0.12 [-0.24, -0.00]	0.18 [0.06, 0.31]	0.02 [-0.11, 0.14]	0.14 [0.01, 0.28]	0.02 [-0.10, 0.14]	0.25 [0.13, 0.38]	–
HUST	77	7.11	-0.02 [-0.14, 0.11]	0.30 [0.19, 0.43]	0.52 [0.42, 0.64]	0.18 [0.06, 0.31]	0.11 [0.01, 0.22]	0.67 [0.57, 0.80]	0.59 [0.48, 0.70]
MATR	169	6.94	-0.27 [-0.46, -0.05]	0.32 [0.17, 0.51]	0.34 [0.17, 0.54]	0.24 [0.07, 0.44]	-0.03 [-0.21, 0.19]	0.61 [0.42, 0.79]	0.29 [0.09, 0.49]

Table 20. Squeeze of the ridge HI: monotonicity and certified information (nats).

Subset		$\gamma = 0.0$	$\gamma = 0.5$	$\gamma = 0.9$	$\gamma = 0.97$	$\gamma = 0.99$	$\gamma = 0.995$	Mon_∞ (predicted)
FD001	Mon	0.24	0.34	0.54	0.56	0.55	0.55	0.55
	certificate	0.31	0.28	0.21	0.09	-0.06	-0.08	
FD002	Mon	0.19	0.28	0.54	0.58	0.58	0.58	0.61
	certificate	0.21	0.20	0.18	0.11	0.00	-0.06	
FD003	Mon	0.16	0.22	0.40	0.41	0.42	0.42	0.41
	certificate	0.10	0.10	0.10	0.07	0.02	-0.07	
FD004	Mon	0.13	0.18	0.36	0.41	0.41	0.41	0.41
	certificate	0.17	0.18	0.16	0.13	0.07	0.00	
HUST	Mon	0.32	0.53	0.80	0.81	0.81	0.81	0.82
	certificate	0.30	0.28	0.22	0.12	0.04	-0.00	
MATR	Mon	0.39	0.46	0.50	0.48	0.48	0.48	0.49
	certificate	0.32	0.28	0.17	-0.02	-0.14	-0.20	

Table 21. Direction of a scalar linear HI: range of stage-optimal angles, best static angle, and worst-stage RMSE ratios to the free direction.

Subset	stages	optimal angle [deg]	static angle [deg]	static / free	level only / free	velocity only / free
FD001	life fraction	43–70	70	1.019	1.18	1.07
FD001	age	33–69	60	1.021	1.20	1.18
FD002	life fraction	18–66	55	1.002	1.13	1.05
FD002	age	21–56	45	1.037	1.19	1.29
FD003	life fraction	11–33	20	1.003	1.05	1.16
FD003	age	6–35	25	1.025	1.08	1.38
FD004	life fraction	1–20	5	1.007	1.02	1.10
FD004	age	9–37	15	1.018	1.05	1.33
HUST	life fraction	3–80	70	1.090	1.13	1.10
HUST	age	0–57	20	1.184	1.08	1.54
MATR	life fraction	42–88	65	1.000	1.06	1.01
MATR	age	0–10	0	1.012	1.01	1.42

Table 22. RMSE of a linear read-out of the life fraction. Blind: without normalisation by regime (engines) or by the cell's own early cycles (cells).

Subset	Regimes	Blind	Normalised	Regime-specific weights
FD001	1	0.139	0.139	0.139
FD002	6	0.148	0.137	0.136
FD003	1	0.156	0.156	0.156
FD004	6	0.199	0.165	0.164
HUST	one per cell	0.136	0.115	–
MATR	one per cell	0.179	0.142	–

Table 23. Age-conditional certificates (nats) of the benchmark methods on the official test engines. †: the age-only control fails.

Method	FD001	FD002	FD003†	FD004
$\bar{h}(T \mid t)$	5.09	5.07	5.61	5.47
Age law (control)	0.07 [-0.04, 0.20]	0.01 [-0.07, 0.09]	0.33 [0.23, 0.44]	0.05 [-0.02, 0.13]
PCA-HI	0.59 [0.49, 0.72]	0.29 [0.21, 0.39]	0.67 [0.54, 0.85]	0.26 [0.19, 0.35]
Geo-opt HI	0.58 [0.48, 0.71]	0.23 [0.13, 0.36]	1.10 [0.94, 1.30]	0.66 [0.55, 0.76]
Ridge HI	0.41 [0.26, 0.58]	0.27 [0.17, 0.39]	0.90 [0.77, 1.05]	0.62 [0.54, 0.71]
β-VAE	0.59 [0.47, 0.72]	0.24 [0.16, 0.34]	0.49 [0.35, 0.65]	0.22 [0.14, 0.31]
VIB	0.15 [0.01, 0.32]	0.06 [-0.01, 0.14]	0.57 [0.45, 0.74]	0.11 [0.02, 0.22]
PGIB, integrating (10 seeds)	0.14 [-0.02, 0.32]	0.07 [-0.01, 0.16]	0.49 [0.37, 0.63]	0.19 [0.08, 0.30]
PGIB-KF (10 seeds)	0.61 [0.45, 0.82]	0.28 [0.18, 0.37]	1.04 [0.87, 1.22]	0.48 [0.39, 0.56]
PGIB-KFR (10 seeds)	0.43 [0.28, 0.62]	0.28 [0.20, 0.37]	0.84 [0.73, 0.98]	0.59 [0.51, 0.69]

References

[1] Si X-S, Wang W, Hu C-H and Zhou D-H 2011 Remaining useful life estimation – a review on the statistical data driven approaches *Eur. J. Oper. Res.* **213** 1–14 doi:10.1016/j.ejor.2010.11.018

[2] Lei Y, Li N, Guo L, Li N, Yan T and Lin J 2018 Machinery health prognostics: a systematic review from data acquisition to RUL prediction *Mech. Syst. Signal Process.* **104** 799–834 doi:10.1016/j.ymssp.2017.11.016

[3] Javed K, Gouriveau R and Zerhouni N 2017 State of the art and taxonomy of prognostics approaches, trends of prognostics applications and open issues towards maturity at different technology readiness levels *Mech. Syst. Signal Process.* **94** 214–36 doi:10.1016/j.ymssp.2017.01.050

[4] Zio E 2022 Prognostics and health management (PHM): where are we and where do we (need to) go in theory and practice *Reliab. Eng. Syst. Saf.* **218** 108119 doi:10.1016/j.ress.2021.108119

[5] Liao L and Köttig F 2014 Review of hybrid prognostics approaches for remaining useful life prediction of engineered systems, and an application to battery life prediction *IEEE Trans. Reliab.* **63** 191–207 doi:10.1109/TR.2014.2299152

[6] Li X, Ding Q and Sun J-Q 2018 Remaining useful life estimation in prognostics using deep convolution neural networks *Reliab. Eng. Syst. Saf.* **172** 1–11 doi:10.1016/j.ress.2017.11.021

[7] Fink O, Wang Q, Svensén M, Dersin P, Lee W-J and Ducoffe M 2020 Potential, challenges and future directions for deep learning in prognostics and health management applications *Eng. Appl. Artif. Intell.* **92** 103678 doi:10.1016/j.engappai.2020.103678

[8] Whitmore G A 1995 Estimating degradation by a Wiener diffusion process subject to measurement error *Lifetime Data Anal.* **1** 307–19 doi:10.1007/BF00985762

[9] Lee M-L T and Whitmore G A 2006 Threshold regression for survival analysis: modeling event times by a stochastic process reaching a boundary *Stat. Sci.* **21** 501–13 doi:10.1214/088342306000000330

[10] Wang L, Chen S, Cai Z, Liu W and Ye Z 2026 Cross-cycle health indicator construction and state of health prediction of lithium-ion batteries by cross recurrence quantification analysis *J. Reliab. Sci. Eng.* **2** 035302 doi:10.1088/3050-2454/ae8dd1

[11] Gu H, Li Y, Jiang L and Luo Z 2025 A generalized approach for remaining useful life prediction considering the multi-source uncertainty of mechanical systems *J. Reliab. Sci. Eng.* **1** 025301 doi:10.1088/3050-2454/add5fa

[12] Bisoyi S, Kumar Rathi A and Mahato S 2026 Reliability assessment of remaining useful life predictions of roller bearings using deep learning models *J. Reliab. Sci. Eng.* **2** 025302 doi:10.1088/3050-2454/ae61b4

[13] Aven T 2021 The reliability science: its foundation and link to risk science and other sciences *Reliab. Eng. Syst. Saf.* **215** 107863 doi:10.1016/j.ress.2021.107863

[14] Sun C-P 2025 Reliability science and engineering: a catalyst for engineering innovation *J. Reliab. Sci. Eng.* **1** 020201 doi:10.1088/3050-2454/adbf22

[15] Coble J and Hines J W 2009 Identifying optimal prognostic parameters from data: a genetic algorithms approach *Annu. Conf. of the Prognostics and Health Management Society* **1**

[16] Javed K, Gouriveau R, Zerhouni N and Nectoux P 2013 A feature extraction procedure based on trigonometric functions and cumulative descriptors to enhance prognostics modeling *Proc. IEEE Conf. on Prognostics and Health Management* pp 1–7 doi:10.1109/ICPHM.2013.6621413

[17] Zhang B, Zhang L and Xu J 2016 Degradation feature selection for remaining useful life prediction of rolling element bearings *Qual. Reliab. Eng. Int.* **32** 547–54 doi:10.1002/qre.1771

[18] Qian F and Niu G 2015 Remaining useful life prediction using ranking mutual information based monotonic health indicator *Proc. Prognostics and System Health Management Conf.* pp 1–5 doi:10.1109/PHM.2015.7380042

[19] Guo L, Li N, Jia F, Lei Y and Lin J 2017 A recurrent neural network based health indicator for remaining useful life prediction of bearings *Neurocomputing* **240** 98–109 doi:10.1016/j.neucom.2017.02.045

[20] Kingma D P and Welling M 2014 Auto-encoding variational Bayes *Proc. Int. Conf. on Learning Representations* arXiv:1312.6114

[21] Higgins I, Matthey L, Pal A, Burgess C, Glorot X, Botvinick M, Mohamed S and Lerchner A 2017 β-VAE: learning basic visual concepts with a constrained variational framework *Proc. Int. Conf. on Learning Representations*

[22] Severson K A et al 2019 Data-driven prediction of battery cycle life before capacity degradation *Nat. Energy* **4** 383–91 doi:10.1038/s41560-019-0356-8

[23] Saxena A, Celaya J, Balaban E, Goebel K, Saha B, Saha S and Schwabacher M 2008 Metrics for evaluating performance of prognostic techniques *Proc. Int. Conf. on Prognostics and Health Management* pp 1–17 doi:10.1109/PHM.2008.4711436

[24] Folks J L and Chhikara R S 1978 The inverse Gaussian distribution and its statistical application – a review *J. R. Stat. Soc. B* **40** 263–75 doi:10.1111/j.2517-6161.1978.tb01039.x

[25] Chhikara R S and Folks J L 1989 *The Inverse Gaussian Distribution: Theory, Methodology, and Applications* (New York: Marcel Dekker)

[26] Tweedie M C K 1957 Statistical properties of inverse Gaussian distributions. I *Ann. Math. Stat.* **28** 362–77 doi:10.1214/aoms/1177706964

[27] Tang S, Yu C, Wang X, Guo X and Si X 2014 Remaining useful life prediction of lithium-ion batteries based on the Wiener process with measurement error *Energies* **7** 520–47 doi:10.3390/en7020520

[28] Gebraeel N Z, Lawley M A, Li R and Ryan J K 2005 Residual-life distributions from component degradation signals: a Bayesian approach *IIE Trans.* **37** 543–57 doi:10.1080/07408170590929018

[29] Wang X 2010 Wiener processes with random effects for degradation data *J. Multivariate Anal.* **101** 340–51 doi:10.1016/j.jmva.2008.12.007

[30] Si X-S, Wang W, Hu C-H, Zhou D-H and Pecht M G 2012 Remaining useful life estimation based on a nonlinear diffusion degradation process *IEEE Trans. Reliab.* **61** 50–67 doi:10.1109/TR.2011.2182221

[31] Medjaher K, Tobon-Mejia D A and Zerhouni N 2012 Remaining useful life estimation of critical components with application to bearings *IEEE Trans. Reliab.* **61** 292–302 doi:10.1109/TR.2012.2194175

[32] Ye Z-S and Xie M 2015 Stochastic modelling and analysis of degradation for highly reliable products *Appl. Stoch. Models Bus. Ind.* **31** 16–32 doi:10.1002/asmb.2063

[33] Zhang Z, Si X, Hu C and Lei Y 2018 Degradation data analysis and remaining useful life estimation: a review on Wiener-process-based methods *Eur. J. Oper. Res.* **271** 775–96 doi:10.1016/j.ejor.2018.02.033

[34] Sankararaman S and Goebel K 2013 Why is the remaining useful life prediction uncertain? *Annu. Conf. of the Prognostics and Health Management Society* **5** doi:10.36001/phmconf.2013.v5i1.2263

[35] Sankararaman S 2015 Significance, interpretation, and quantification of uncertainty in prognostics and remaining useful life prediction *Mech. Syst. Signal Process.* **52–53** 228–47 doi:10.1016/j.ymssp.2014.05.029

[36] Hüllermeier E and Waegeman W 2021 Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods *Mach. Learn.* **110** 457–506 doi:10.1007/s10994-021-05946-3

[37] Kendall A and Gal Y 2017 What uncertainties do we need in Bayesian deep learning for computer vision? *Advances in Neural Information Processing Systems* **30** arXiv:1703.04977

[38] Lakshminarayanan B, Pritzel A and Blundell C 2017 Simple and scalable predictive uncertainty estimation using deep ensembles *Advances in Neural Information Processing Systems* **30** arXiv:1612.01474

[39] Shannon C E 1948 A mathematical theory of communication *Bell Syst. Tech. J.* **27** 379–423, 623–56 doi:10.1002/j.1538-7305.1948.tb01338.x

[40] Cover T M and Thomas J A 2006 *Elements of Information Theory 2nd edn* (Hoboken, NJ: Wiley-Interscience)

[41] Tishby N, Pereira F C and Bialek W 2000 The information bottleneck method arXiv:physics/0004057

[42] Alemi A A, Fischer I, Dillon J V and Murphy K 2017 Deep variational information bottleneck *Proc. Int. Conf. on Learning Representations* arXiv:1612.00410

[43] Poole B, Ozair S, van den Oord A, Alemi A A and Tucker G 2019 On variational bounds of mutual information *Proc. Int. Conf. on Machine Learning (PMLR vol 97)* pp 5171–80

[44] Achille A and Soatto S 2018 Emergence of invariance and disentanglement in deep representations *J. Mach. Learn. Res.* **19** 1–34

[45] Kolchinsky A, Tracey B D and Wolpert D H 2019 Nonlinear information bottleneck *Entropy* **21** 1181 doi:10.3390/e21121181

[46] Saxe A M, Bansal Y, Dapello J, Advani M, Kolchinsky A, Tracey B D and Cox D D 2019 On the information bottleneck theory of deep learning *J. Stat. Mech.: Theory Exp.* **2019** 124020 doi:10.1088/1742-5468/ab3985

[47] Vasicek O 1976 A test for normality based on sample entropy *J. R. Stat. Soc. B* **38** 54–9 doi:10.1111/j.2517-6161.1976.tb01566.x

[48] Kraskov A, Stögbauer H and Grassberger P 2004 Estimating mutual information *Phys. Rev. E* **69** 066138 doi:10.1103/PhysRevE.69.066138

[49] Paninski L 2003 Estimation of entropy and mutual information *Neural Comput.* **15** 1191–253 doi:10.1162/089976603321780272

[50] McAllester D and Stratos K 2020 Formal limitations on the measurement of mutual information *Proc. Int. Conf. on Artificial Intelligence and Statistics (PMLR vol 108)* pp 875–84

[51] Van Trees H L 1968 *Detection, Estimation, and Modulation Theory, Part I* (New York: Wiley)

[52] Gill R D and Levit B Y 1995 Applications of the van Trees inequality: a Bayesian Cramér–Rao bound *Bernoulli* **1** 59–79 doi:10.2307/3318681

[53] Tichavský P, Muravchik C H and Nehorai A 1998 Posterior Cramér–Rao bounds for discrete-time nonlinear filtering *IEEE Trans. Signal Process.* **46** 1386–96 doi:10.1109/78.668800

[54] Anonymous 2026 *Reference withheld for double-blind review*

[55] Anonymous 2026 *Reference withheld for double-blind review*

[56] Anonymous 2026 *Reference withheld for double-blind review*

[57] Bishop C M 2006 *Pattern Recognition and Machine Learning* (New York: Springer)

[58] Kalman R E 1960 A new approach to linear filtering and prediction problems *J. Basic Eng.* **82** 35–45 doi:10.1115/1.3662552

[59] Durbin J and Koopman S J 2012 *Time Series Analysis by State Space Methods 2nd edn* (Oxford: Oxford University Press)

[60] Stam A J 1959 Some inequalities satisfied by the quantities of information of Fisher and Shannon *Inf. Control* **2** 101–12 doi:10.1016/S0019-9958(59)90348-1

[61] Durrett R 2010 *Probability: Theory and Examples 4th edn* (Cambridge: Cambridge University Press)

[62] Harvey A C 1989 *Forecasting, Structural Time Series Models and the Kalman Filter* (Cambridge: Cambridge University Press)

[63] Genest C and Zidek J V 1986 Combining probability distributions: a critique and an annotated bibliography *Stat. Sci.* **1** 114–35 doi:10.1214/ss/1177013825

[64] Hinton G E 2002 Training products of experts by minimizing contrastive divergence *Neural Comput.* **14** 1771–800 doi:10.1162/089976602760128018

[65] Meeker W Q and Escobar L A 1998 *Statistical Methods for Reliability Data* (New York: Wiley)

[66] Gneiting T and Raftery A E 2007 Strictly proper scoring rules, prediction, and estimation *J. Am. Stat. Assoc.* **102** 359–78 doi:10.1198/016214506000001437

[67] Bouthillier X et al 2021 Accounting for variance in machine learning benchmarks *Proc. Mach. Learn. Syst.* **3** 747–69

[68] Efron B and Tibshirani R J 1993 *An Introduction to the Bootstrap* (New York: Chapman and Hall)

[69] Ma G et al 2022 Real-time personalized health status prediction of lithium-ion batteries using deep transfer learning *Energy Environ. Sci.* **15** 4083–94 doi:10.1039/D2EE01676A

[70] Attia P M et al 2020 Closed-loop optimization of fast-charging protocols for batteries with machine learning *Nature* **578** 397–402 doi:10.1038/s41586-020-1994-5

[71] Tan R et al 2025 BatteryLife: a comprehensive dataset and benchmark for battery life prediction *Proc. 31st ACM SIGKDD Conf. on Knowledge Discovery and Data Mining* **2** pp 5789–800 doi:10.1145/3711896.3737372

[72] Saha B and Goebel K 2007 Battery data set *NASA Prognostics Data Repository (Moffett Field, CA: NASA Ames Research Center)*

[73] He W, Williard N, Osterman M and Pecht M 2011 Prognostics of lithium-ion batteries based on Dempster–Shafer theory and the Bayesian Monte Carlo method *J. Power Sources* **196** 10314–21 doi:10.1016/j.jpowsour.2011.08.040

[74] Nectoux P, Gouriveau R, Medjaher K, Ramasso E, Chebel-Morello B, Zerhouni N and Varnier C 2012 PRONOSTIA: an experimental platform for bearings accelerated degradation tests *Proc. IEEE Int. Conf. on Prognostics and Health Management (Denver, CO)* pp 1–8

[75] Lee J, Qiu H, Yu G, Lin J and Rexnord Technical Services 2007 Bearing data set *IMS, University of Cincinnati, NASA Prognostics Data Repository (Moffett Field, CA: NASA Ames Research Center)*

[76] Qiu H, Lee J, Lin J and Yu G 2006 Wavelet filter-based weak signature detection method and its application on rolling element bearing prognostics *J. Sound Vib.* **289** 1066–90 doi:10.1016/j.jsv.2005.03.007

[77] Saxena A, Goebel K, Simon D and Eklund N 2008 Damage propagation modeling for aircraft engine run-to-failure simulation *Proc. Int. Conf. on Prognostics and Health Management (Denver, CO)* pp 1–9 doi:10.1109/PHM.2008.4711414

[78] Welch B L 1947 The generalization of 'Student's' problem when several different population variances are involved *Biometrika* **34** 28–35 doi:10.1093/biomet/34.1-2.28